

STARTER-GENERATOR

200SGL Series

OVERHAUL INSTRUCTIONS

with

ILLUSTRATED PARTS BREAKDOWN

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Keep this record in the front of manual. When you get revisions, replace the affected pages in manual with the revised pages. On this page, write the revision number, date issued, date inserted, and incorporator's initials.							

SERVICE BULLETIN LIST

These Service Bulletins are applicable to this manual.

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132	Dec 16/98	199	Mar 15/08
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159	Oct 29/02		
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SECTION I
INTRODUCTION

1. General.

- A. This manual contains maintenance instructions, test procedure and Illustrated Parts Breakdown (IPB) for type 200SGL Series DC Starter-Generator. This unit is manufactured by Aircraft Parts Corporation, 100 Corporate Drive, Holtsville, NY 11742, U.S.A
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All technical documentation and information contained herein, with respect to Assembly and Disassembly, Cleaning, Inspection methods and limits, Repair methods and limits, Operational limits, life limits and the like, have been developed and approved for use with parts that have been manufactured and/or approved by Aircraft parts Corporation and that have been maintained in accordance with APC technical documentation and recommendations. APC has no contractual or legal obligation for, nor knowledge of, non-APC-approved parts and repairs. Accordingly, this document is not intended to apply to non-APC-approved parts or repairs.

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 - Read and obey specified instructions in MSDS for use and types of personal protective equipment (safety glasses, gloves, aprons, etc.), ventilators or respirators, fire extinguishers, and procedures for a medical emergency.
 - Read and obey the hazardous materials label on the container of hazardous material, and the MSDS supplied by the manufacturer.
 - Obey standard shop practices and procedures when you use hazardous materials, and when you put hazardous materials in storage.
 - When you discard hazardous materials, obey all federal, state, and local regulations.
- E. Values in this manual are given in US Standard Units, followed in parentheses by equivalent values for the International System of Units (Metric or SI units).

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SECTION II
DESCRIPTION AND OPERATION

2-1. General.

- A. This manual contains maintenance instructions, test procedure and IPB for type 200SGL Series DC Starter-Generator. Section III of this manual contains maintenance instructions for the 200SGL Series Starter Generator. Test procedure is covered in Section IV. Section V contains an IPB.

2-2. Description.

- A. The Starter-Generator is used as an engine-driven source of DC power in a 28 volt aircraft electrical system and as a source of starting power for lightweight turbine engines. An efficient integral fan draws air in at anti-drive end of generator to permit self-cooled operation.

2-3. Specifications.

Table 2-1
Specifications

DC Output	
Volts	30 ± 0.1
Amperes	200 ± 5
Speed Range	
Max Speed (rpm)	13,000 +0/-100
Continuous Operating Speed (rpm)	12,100 ± 100
Minimum rated Speed (rpm)	7,050
Overspeed (rpm)	14,000 ± 100
Cooling	Self-Cooled
Rotation (Viewed from Drive end)	CCW
Spline Data	
Teeth	12
Pitch	20 / 40
Pressure Angle	30°
Mounting Flange	Quick-Attach-Detach (QAD) MS3331-1P
Overall Dimensions	
Diameter (max.) Past Terminal Block	5.41
Length (max.)	9.10
External Starting Power	
Volts, DC	28.5 Max.
Amperes	1000 Max.

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2-4. Applicable Part Numbers

This overhaul manual provides coverage for the following part numbers:

200SGL111Q	200SGL119Q-2	200SGL130Q	200SGL144Q
200SGL111Q-2	200SGL120Q	200SGL130Q-2	200SGL145Q
200SGL111Q-3	200SGL121Q	200SGL130Q-3	200SGL147Q
200SGL112Q	200SGL124Q	200SGL132Q	200SGL147Q-2
200SGL114Q	200SGL129Q	200SGL132Q	200SGL147Q-3
200SGL115Q	200SGL129Q(1)	200SGL132Q-2	200SGL150Q
200SGL115Q-2	200SGL129Q(3)	200SGL132Q-3	200SGL150Q-1
200SGL117Q-2	200SGL129Q(4)	200SGL132Q-4	200SGL150Q-2
200SGL118Q	200SGL129Q(5)	200SGL142Q	200SGL151Q
200SGL119Q	200SGL129Q(8)		

NOTE: This list shows all part numbers covered by this manual. In some cases, model numbers have been superceded by higher Q- numbers (example: 200SGL119QQ has been superceded by 200SGL119Q-1).

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SECTION III
MAINTENANCE INSTRUCTIONS

3-1. Special Tools

3-2. Special tools and test equipment are listed in Table 3-1.

Table 3-1
Special Tools and Test Equipment

Part Number	Nomenclature	Application
NOTE: Equivalent substitutes may be used for the items listed below.		
TUSG100170-3	Base Support Plate for 200SGL Series	Base Plate.
T150SG1073-1	Tool, Damper Plate	Position Armature Damper plate on Armature shaft.
TUSG121900-11	Spur Gear Installation Tool	Pressing on spur gear
T150SG1052-2	Bearing Pusher	Pressing on bearings
T150SG1052-3	Bearing Pusher	Pressing on bearings
	Growler	To test intercoil shorts in armatures
	Hi- Pot Tester	To test for shorts in electrical wiring.
	Nano Volt/Ohm Meter or equivalent	For measuring resistance of various coils.
	Capacitance Tester	Testing of capacitors
	Dynamic Balancer	For checking armature balance.

3-3. Disassembly

3-4. General

A. Disassemble the Starter-Generator as shown in parts list (Figure 5-1 through 5-3).

NOTE: Retaining ring, O-ring, bearings, lock washers, flat washers, screws, and self locking nuts must be discarded whenever removed. Replace with new hardware. Brush screws and terminal block screws may be re-used if in good condition.

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3-5. Disassembly (Refer to IPL Figure 5-1 & 5-2).

- A. Remove fan cover. Lift up brush springs (Figure 5-3) and remove and discard brushes.

NOTE: 200SGL "XL" Series Starter Generators: Based on the established brush life for the operator whose generator is being serviced, and the Time Between Overhauls (TBO), the operator can amend this paragraph to utilize the brushes for another operating period. If this is the case, the brushes and brush holders should be marked so as to return the same brush to the brush box it was removed from. This will reduce time to re-seat the brushes. Brush life remaining can be calculated as per "Brush Life Calculation Procedure" at the end of this section.

- B. Hold the drive end of drive shaft assembly using a spline wrench. Remove lock nut, springs and fan. Remove the drive shaft assembly.
- C. Remove back damper plate using gear puller, taking care to prevent damage to end of armature shaft by gear puller.
- D. Remove four screws and carefully lift connector from spacer. Unsolder lead wire from connector pins and remove connector and spacer from stator housing assembly, if applicable.
- E. Remove eight screws and washers. Remove Drive End Bell.
- F. Remove eight screws, lock washers and washers. Remove commutator end bell assembly. Press out armature using a suitable pusher to protect armature shaft end.
- G. Remove spur gear (18) if applicable. Remove retaining ring and baffle disc, then using a bearing puller, remove ball bearing from generator armature. Remove 4 screws from bearing retainer. Press ball bearing out of commutator head. Do not disassemble commutator head unless necessary.
- H. Do not remove the magnetic pick-up or change its settings unless necessary.
- I. Do not disassemble stator assy. To remove terminal block, remove nuts and washers from terminal block lugs, and remove screws and washers that connect terminal block to stator. Then remove terminal block.

* 3-6. General Cleaning

- A. The following Warning and Cautions apply to cleaning the starter generator.

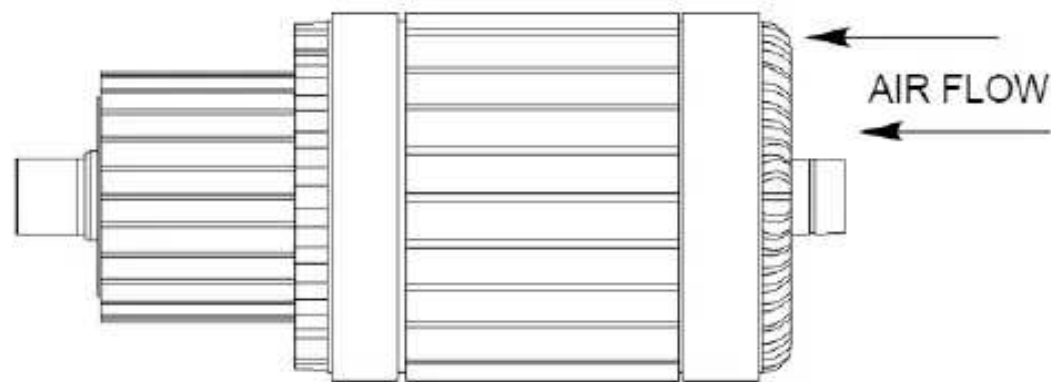
WARNING: WHEN USING CLEANING SOLVENTS, TAKE PRECAUTION TO AVOID BURNING SKIN OR INHALING FUMES. USE SOLVENT IN A WELL-VENTILATED ROOM ONLY.

WARNING: PUT ON APPROVED SAFETY EQUIPMENT TO PREVENT INJURY TO EYES AND FACE WHEN USING COMPRESSED AIR. MAXIMUM PERMITTED AIR PRESSURE IS 30 PSIG (207 KPA, GAGE). DO NOT POINT AIR OUTLET AT YOURSELF OR OTHER PERSONS.

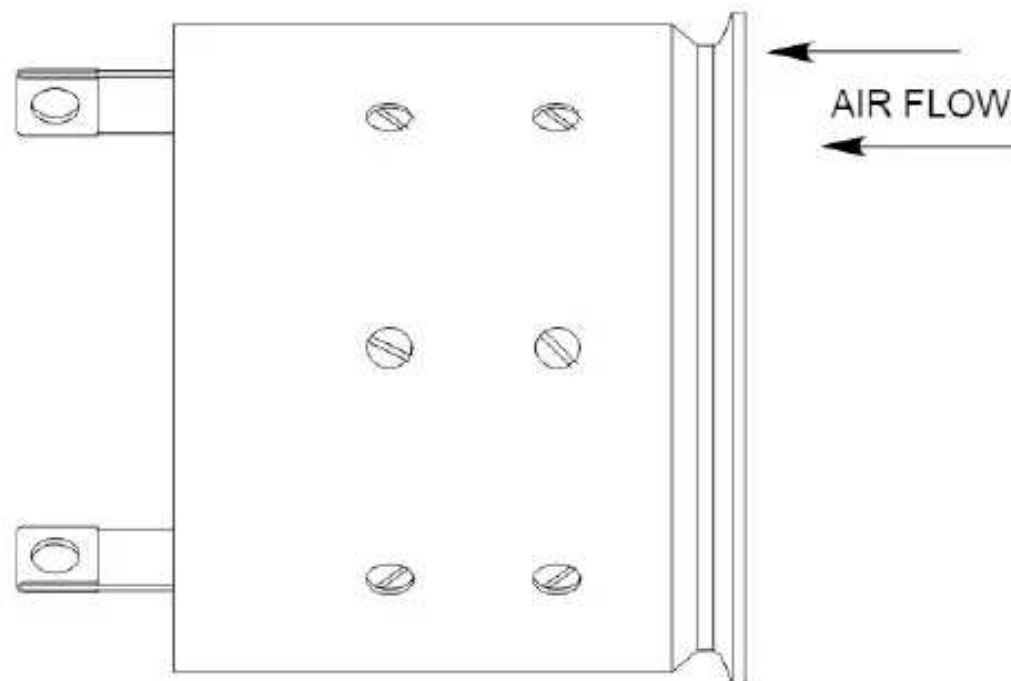
CAUTION: SOLVENTS OR CLEANING METHODS OTHER THAN THOSE DESCRIBED ABOVE, ARE NOT RECOMMENDED SINCE THEIR USE MAY DAMAGE THE INSULATING VARNISH USED IN THIS EQUIPMENT.

CAUTION: DO NOT APPLY OIL TO THE COMMUTATOR OR ELECTRICAL WIRES OR

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Cleaning of Armature
Figure 3-1



Cleaning of Stator
Figure 3-2

- B. After the Starter Generator is disassembled, blow off the parts with compressed air to remove carbon dust. When using air to remove contaminants from Armature and Stator, blow air in direction as shown in Figure 3-1 and 3-2.
- C. Clean components with Safety Kleen premium gold solvent or equivalent and stiff non-wire brush.
- D. Place components in JRI parts washer or equivalent, set the washer timer to 30 minutes. The use of a non-conductive detergent is recommended.
- E. Place armature and stator in an ultrasonic cleaning tank for 30 minutes.
- F. After cleaning process, rinse parts with fresh water and place armature and stator in drying ovens for 6 hours at (250°-280° F) to evaporate moisture. Commutator heads should be placed in oven for 6 hours at 150° F max.
- G. Apply a light film of oil or anti-corrosion compound to all bare steel metal surfaces such as bearing liners, armature drive shafts, and damper faces to prevent rusting, if storing for a length of time.

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* 3-7. Inspection

3-8. General.

CAUTION: PLATING OR REWORK OF DRIVE END OR COMMUTATOR END HOUSING BEARING LINERS IS NOT RECOMMENDED. THIS PROCEDURE CAN LEAD TO PREMATURE BEARING FAILURES.

- a. Inspect all parts and wiring for wear and damage. Refer to Table 3-2 for wear inspection measurements.
- b. Replace all worn or damaged parts except where a repair procedure is given.

Table 3-2
Wear Inspection Measurements

See Figure 5-1	Checkpoint	Dimensions/Inches	
		Min	Max
Armature (44)	Armature Bearing Journals OD	0.6692	0.6695
Drive End Bell (10)	Bearing Liner ID	1.5747	1.5750
Commutator Housing (48)	Housing Bore ID	1.5747	1.5750
Armature (44)	Commutator OD	1.9600	-
* Further factory repair may be possible using Aircraft Parts Corporation OEM repair Work Instruction No. 7.5.1.1431.			

* 3-9. Armature (Refer to IPL Figure 5-1)

- a. Examine the armature for wear and damage.
- b. Check the bands of the armature for firmness and tightness. Check the commutator for loose or misaligned bars. Measure the depth of undercut between commutator bars. The depth should not be less than .032 in. (0.812 mm). If so, cut the slot to a depth of .045 in. (1.143 mm) and a width of .030 - .035 in. (0.762 - 0.889 mm). Inspect the brush contact surfaces of commutator, which should be even, highly burnished and copper in color. If brush contact surfaces are rough, pitted, scored, burned, or have a film of carbon or oil that cleaning will not remove, turn down the commutator and undercut as instructed in paragraph 3-21. (It is recommended the commutator be refinished each time brushes are changed.)
- c. Badly burned commutator bars are usually the result of:
 - open or shorted armature coils
 - undersize brushes
 - brushes too tight in brush boxes
 - loose brush terminal screws
 - operating generator in starter mode for too long a time.
 - hung brush

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- d. Inspect the armature windings for burned, cracked, or frayed insulation. Check to see all conductors are firmly brazed to commutator risers.
- e. Closely examine the splines inside the armature shaft for wear. The degree of spline wear may be determined by visual comparison with a new spline. Replace the armature if a new drive shaft shows excessive radial play when inserted in the armature.

3-10. Drive Shaft Assembly (Refer to IPL Figure 4-1)

- a. Check the drive shaft assembly splines for wear. Large 12 tooth splines should measure .696 in. (17.679 mm) minimum over two .0864 in. (2.194 mm) diameter pins. The small 11 tooth spline is checked for wear by installing it in the armature and a new drive shaft into the same armature, and then comparing both for radial play. Discard the used shaft for excess radial play.
- b. Mag Particle Inspect (MPI) the drive shaft, during overhaul, per ASTM-E-1445 for cracks, in particular the shear point. Discard the shaft if a crack is noted.
- c. When inspecting for evidence of overheating, bear in mind that although discoloration may indicate a weakened shaft, the condition can also be present in a sound shaft. Uniform discoloration along the pencil section of a new shaft may be disregarded. However, if localized or blotchy discoloration is noted, overheating in service is indicated. This shaft should be replaced. If slight bluing has occurred, the shaft may still be serviceable provided a magnetic particular inspection reveals no surface cracks. If black discoloration is present, discard the drive shaft assembly. Steps should be taken also to eliminate the cause of the condition, such as, overload or excessive vibration.

*

3-11. Commutator, End Bell Assembly (Refer to IPL Figure 5-3.)

- a. Inspect the parts of the commutator end bell assembly for wear and damage. Be sure that brush holder assemblies are not bent, distorted, or loose.
- b. Using an inside micrometer check the bearing liner for wear and damage. If bearing liner inside diameter is worn beyond permissible dimensions specified in Table 3-2, replace the commutator head.
- c. Using a brush spring scale with a small loop of wire, measure the tension of all brush springs, lifting the spring a maximum of .0156 in. (0.3962 mm), off a new brush which is flush with the bottom of brush box. Tension should be between 30-45 ounces.

*

3-12. Stator (Refer to IPL Figure 5-1 and 5-2)

- a. Inspect the stator for wear and damage. Check for damaged insulation and loose or broken terminals. Check for tightness of coils visible brazed joints, and tightness of welded screens. Screens should not rest on stator coils.

3-12A. Drive End Bell (Refer to IPL Figure 5-1)

- a. Measure the bearing liner in drive end bell with gage as per 3-11 (b) above. Replace end bell if liner is worn beyond those dimensions shown in Table 3-2.

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3-13. Testing

* 3-14. Armature (Refer to IPL Figure 5-1)

NOTE: The rewinding of the armature, replacement of the commutator, or rewinding of the stator is not recommended and will void any warranty.

- a. After cleaning and baking armature per paragraph 3-6 and using a Hi-Pot insulation tester, ground test armature. Apply 250 VAC, 60 Hz for 1 second between a commutator bar and the shaft (ground), between the two stainless steel bands and the commutator, and between the two stainless steel bands and the shaft. Replace the armature if it is grounded, unless re-cleaning and baking for six hours at 250° to 280° F (121° to 138°C) will reduce the short to within the limits stated above.

WARNING: WHEN MAKING ELECTRICAL CHECKS, TAKE PRECAUTIONS TO AVOID TOUCHING CONDUCTORS CARRYING HIGH VOLTAGES. MAKE SURE TEST PROBES ARE WELL INSULATED.

CAUTION: KEEP TEST PROBE OUTSIDE OF COMMUTATOR BRUSH PATH TO AVOID DAMAGING THE CONTACT SURFACE.

- b. Test for a short or open circuit by placing the armature in a growler. Holding a thin strip of steel (such as a hacksaw blade) over the armature core, slowly rotate the armature by hand. If the steel strip vibrates, replace the armature if undercutting does not remove short.

* 3-15. Stator (Refer to IPL Figure 5-1)

- a. Using a Hi-Pot insulation tester, ground test the stator. Apply 250 VAC at commercial frequency (60 Hz) for one second between any winding and the stator housing (ground). Replace the stator if the insulation breaks down.
- b. Using a Nano Volt / Ohm Meter, or equivalent device, measure shunt field (terminals A to D). The resistance of the shunt field should be within acceptable tolerance as listed on applicable data sheet. If the resistances are outside the permissible range, replace the stator.

3-16. Commutator Head (Refer to IPL 5-1)

CAUTION: DO NOT TEST ACROSS ADJACENT BRUSH HOLDERS AS CAPACITORS WIRED ACROSS THE HOLDERS WILL BE DAMAGED.

- a. Test the commutator head for grounds and shorts, using Hi-Pot tester described in Table 3-1. Check for shorts between the end bell casting and brush holders. Apply 250 VAC at commercial frequency (60 Hz) for one second. If ground is indicated, disassemble the end bell to repair the short. (See paragraph 3-26)
- b. If commutator head was manufactured with Printed Circuit Board (PCB) assemblies then capacitances must fall within tolerances listed on applicable data sheets.

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3-17. Capacitors and Terminal Block (Refer to IPL 5-1).

- a. Using a capacitance bridge, measure the capacitance of the capacitors in the terminal block. The capacitance must measure as indicated on applicable data sheets.

3-18. Repair or Replacement

3-19. General

- a. Replace all worn or damaged parts except where a repair procedure is given.
- b. Replace all brushes (Figure 5-3) at each overhaul. New brushes must be run-in for proper seating. (Refer to paragraph 4-10.)
- c. Replace ball bearings (Figure 5-1) at each overhaul.
- d. Replace damper springs (Figure 5-1) and nut at each overhaul. Also replace friction ring at each overhaul.

NOTE: This is important for proper operation of drive shaft damper assembly.

3-20. Armature

- a. Replace the armature (IPL Figure 5-1) if any of the following conditions exist:
 - 1. If the bearing journals diameter are not within .6692-.6695 in. (16.997-17.005 mm) (See Table 3-2).
 - 2. If the commutator bars are badly burned and cannot be repaired by cutting on a lathe. (See paragraph 3-11).
 - 3. If the commutator bars are loose or out of alignment.
 - 4. If turning down the commutator would reduce the outside diameter below the allowable minimum of 1.960 in. (49.784 mm).

* 3-21. Resurfacing the Commutator

The commutator of an XL Series Starter-Generator must be resurfaced by an APC authorized repair station equipped with specialized commutator cutting equipment. Correct cutting will remove .002-.004 in. (.0508-.1016 mm) while holding .0002 in. (.0050 mm) concentricity between the commutator and bearing journals. Minimum commutator diameter is as stated below.

If approved commutator turning equipment is not available, the armature can be returned to APC for immediate turning at the customer's expense.

- a. At each overhaul, cut the commutator to remove wear marks, oil or carbon film. A diamond tipped cutting or a carboloy – tipped commutator cutting tool is recommended (See Figures 3-3 & 3-4). The amount of material to be removed is relative to the condition of the commutator surface. Surface should be smooth and free of damage. The point of the cutting tool must be held on the centerline of the armature shaft. (See Figure 3-5). If a carboloy-tipped commutator cutting tool (See Figure 3-4) is used, the point of the cutting tool must be held .0312 in. (.7924 mm) below the center line of the armature shaft. The minimum diameter to which the commutator may be turned down is 1.960 in.

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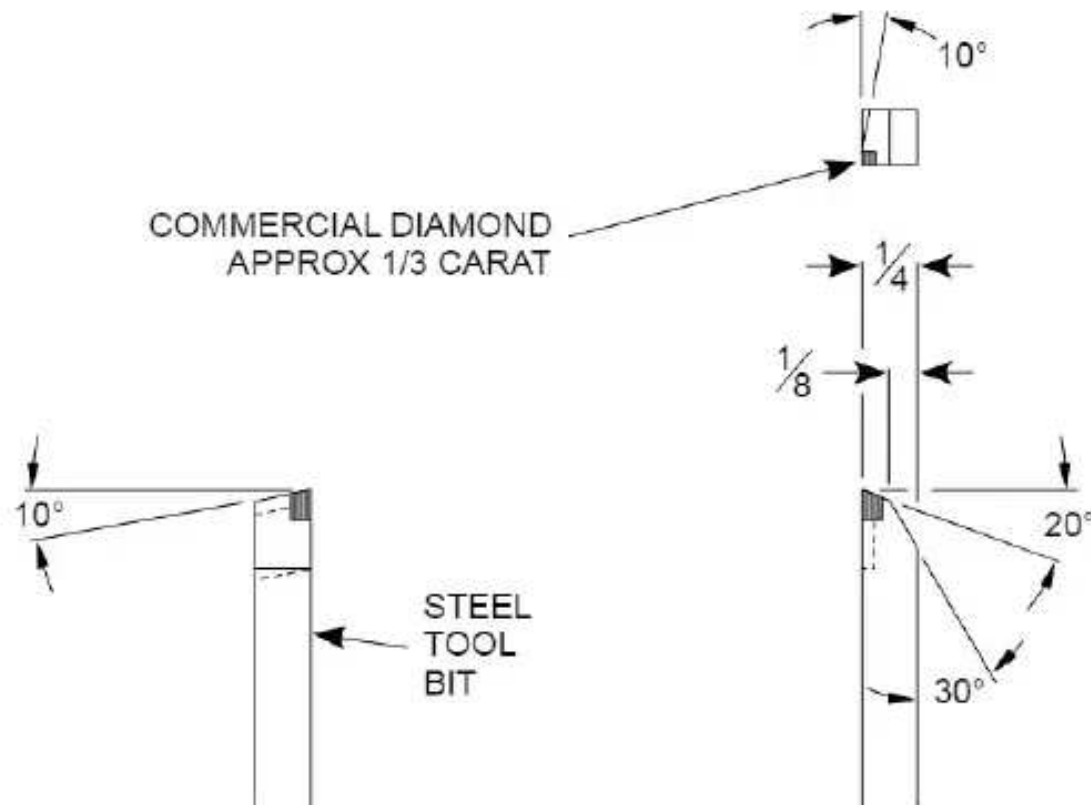


Figure 3-3
Typical Diamond-Tipped Commutator Cutting Tool

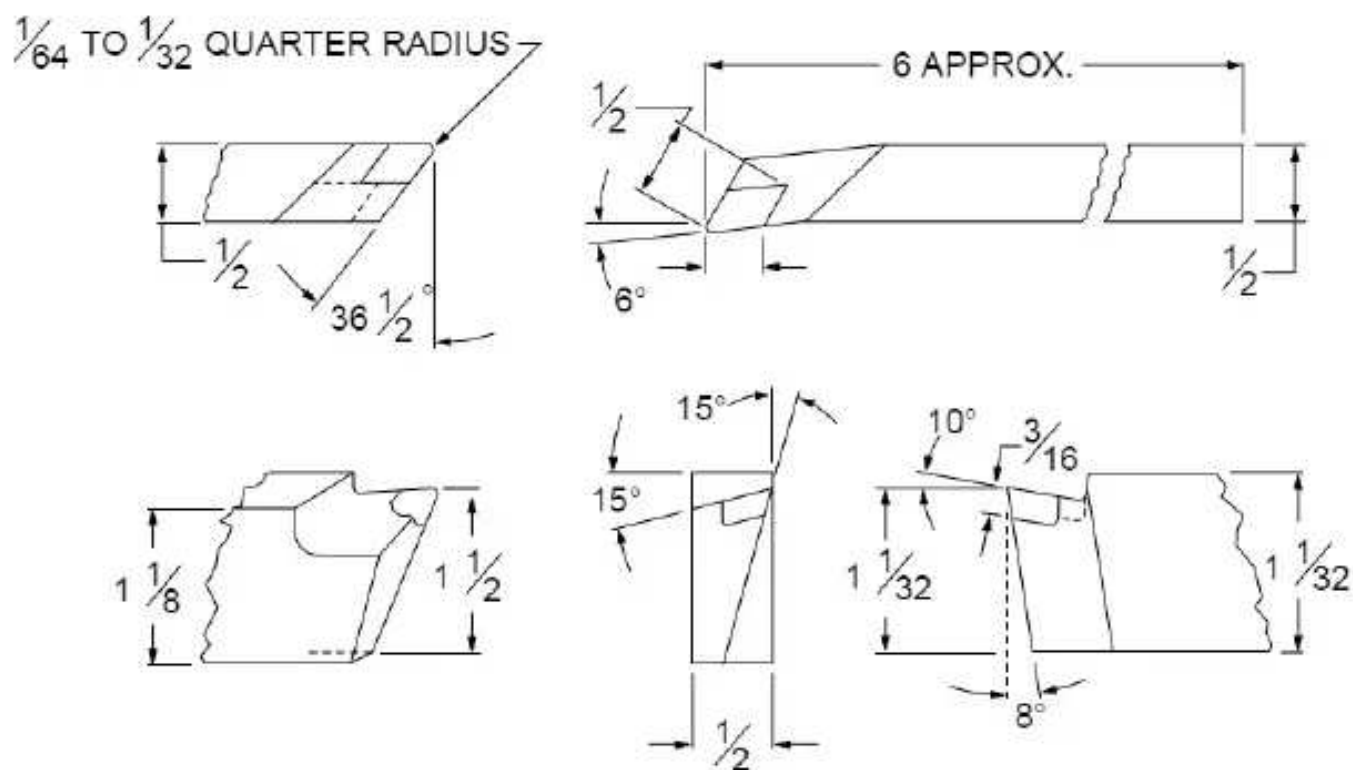


Figure 3-4
Typical Carbide-Tipped Commutator Cutting Tool

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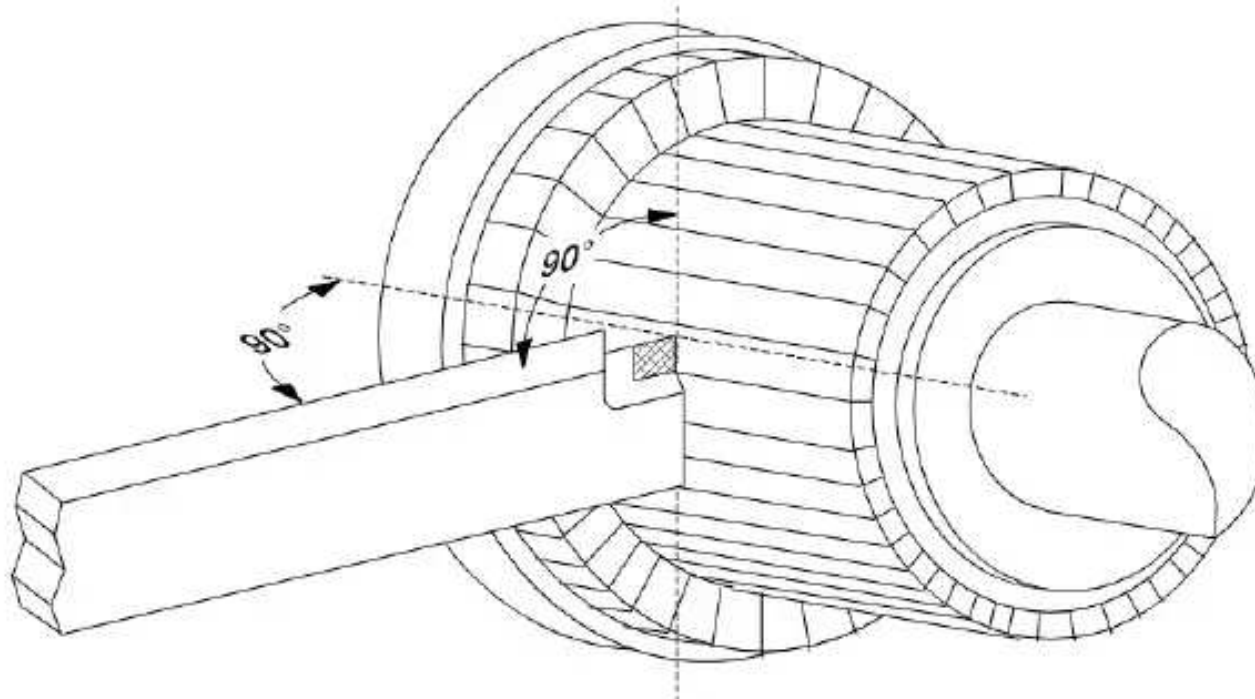


Figure 3-5
Mounting of Commutator Cutting Tool

- b. After the commutator is turned down, measure the depth of the under-cutting between commutator bars. If the depth is less than .032 in. (.8128 mm), undercut the mica to .045 inches deep maximum and to a width of .030-.035 in. (.762-.889 mm).
- c. After the commutator is undercut, take a final cut of not more than 0.001 in. (.0025 mm) across the face of the commutator to remove burrs. If a diamond-tipped commutator cutting tool is not available, make the final cut with a freshly honed carboloy-tipped commutator cutting tool. Do not use polishing abrasives. After the final cut, remove burrs between commutator bars with a strip of fiber or an Exacto knife blade.
- d. The commutator should be concentric with bearing surfaces of the armature shaft within a 0.0005 in. (.0127 mm) full indicator reading. Bar to bar concentricity should be within .0002 (.0050 mm).

3-22. Rebalance of Armature

- a. Using a dynamic balance machine, check the armature for static and dynamic balance. The armature should not be out of balance by more than 5 grain inches. If balance is required, mill the stainless steel bands at either end of armature to a maximum of 0.031 in. (.787 mm) deep max. and 0.250 in. (6.35 mm) wide cuts, until proper balance is obtained. Only two 0.25 in. (6.35 mm) wide cuts are permitted in the width of each band to maintain band strength.

3-23. Commutator End Bell, Bearing Liner and Drive End Bell Bearing Liner (Refer to Figure 5-1)

- a. If the commutator end bell bearing liner or the drive end mounting flange bearing liner are worn beyond 1.5747 to 1.5750 in. (3.9997-4.0005 mm), replace the commutator end bell or drive end bell. Replating of the bearing liner is not recommended.

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3-24. Assembly

3-25. General

- a. Reassembly is the reverse of disassembly.
- b. Follow the exploded views (Figures 5-1 through 5-3), disassembly paragraph 3-3 through 3-5 and the schematic wiring diagram (Figure 3-8) for reassembly information.
- c. Special procedures and tool applications are covered in paragraph 3-29.

3-26. Commutator End Bell Assembly (Refer to Figure 5-3)

NOTE: Paragraphs 3-26 (a) - (e) are for complete break down and repair of Commutator Head.

- a. Place end bell (Figure 5-1) in a suitable holding fixture.

CAUTION: DO NOT OVERWIND SPRING. OVERWINDING SPRING WILL CAUSE SPRING TO TAKE PERMANENT SET AND PRODUCE INSUFFICIENT PRESSURE.

- b. Install two brush springs (Figure 5-3) on brush holder. Each spring must be wound approximately 3/4 turn before installation on brush holder spring support bar.
- c. Install insulating tubes, non-metallic washers, metal washers, and lock nut on the brush box mounting screw in proper order. Position printed circuit board assembly across to adjacent brush holders.
- d. Coat insulating tubes, non-metallic washers and metallic washers prior to installation with red insulator No. ER-41 and thinner T-200X (John C. Dolph Co., Monmouth Junction, NJ). Allow to dry for 30 minutes.
- e. Using a wrench, tighten nut until snug. Torque Nut to 25-30 in. lbs (2.824-3.389 N-m).
- f. Coat area between nut and screw with ER/41 or insulating paint to prevent entry of carbon dust under washers and brush box mounting ears. This will prevent shorting to ground of brush boxes. Bake end bell for two hours.

3-27. Main Assembly (Refer to IPL Figure 5-1)

- a. Mount baffle disc with protruding surface facing bearing on the drive end of generator armature. Mount bearing on same end of armature shaft using an aluminum block or suitable fixture to support armature shaft on opposite end. Press on bearing by pushing the inner race with a suitable bearing pusher. See Fig. 3-7.
- b. Apply Anti-Seize compound part number 150SG1309 or equivalent to commutator end bell bearing race. Press bearing into bell assy, commutator end, using a suitable bearing pusher against the outer race. Secure bearing in end bell with bearing retainer using four screws, and lock washers. Apply anti-seize compound part number 150SG1309 to all four screws.

NOTE: If unit is equipped with capacitor board leads that connect to bearing retainer leave unattached until unit is hi-pot tested at the end of assembly. If unit passes final hi-pot before functional testing connect the two leads. Installation is made

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- c. Apply anti-seize compound part number 150SG1309 to threads of eight screws. Install eight lock washers and eight plain washers on screws. Now install comm end bell to stator assembly and insert screws and tighten. Position commutator end bell and stator assembly, on suitable support fixture, comm end down and insert armature into comm end bell bearing. Use arbor press and suitable bearing pusher to ensure that armature is fully seated in comm end bell bearing.
- d. Apply a continuous film of Never-Seize Anti-Seize compound to the bearing liner of the drive end bell assembly, then position the drive end bell assembly over drive end bearing.

NOTE: If unit is equipped with speed sensor on drive end bell route two speed sensor wires thru hole in stator assembly.

- e. Insure that the bell is fully seated against stator housing. Apply anti-seize compound part number 150SG1309 to threads of screws. Place lock washers and plain washers on screws. Install eight screws.
- f. Install remaining baffle disc on shaft with protruding surface facing bearing. Install retaining ring in groove in armature shaft. Make sure retaining ring is fully seated in groove.

NOTE: Some models (equipped with spur gear and magnetic pickup) will require the additional installation procedures. Refer to IPL and note below for installation differences.

- g. Assemble speed pick-up and spur gear.

If grommet was removed, install grommet. If magnetic pick-up was removed at disassembly, install pick-up loosely into thread boss on drive end bell.

- 1) If heat shrinkable sleeving was removed at disassembly, cover leads above pick-up with 1/8" diameter heat shrinkable sleeving MIL-I-23053-15. Use heat gun to shrink sleeving.
- 2) Feed end of pick-up leads through hole in stator, making sure the leads are out of the way of the armature.
- 3) Press spur gear (Figure 5-2) on armature shaft with small extended hub against ball bearing.
- 4) If pick-up was removed from DE end bell, adjust pick-up to obtain 0.004 to 0.006 in. (.102-.152 mm) gap between pick-up and spur gear. Tighten locknut on magnetic pick-up and recheck gap again. Using lacing tape MIL-T-43435, tie pick-up leads to screen.
- 5) Pull leads gently through hole in stator hole in the spacer. When correctly installed, lead groove in spacer will be against stator housing.
- 6) Slide #12 Varflex A397, Grade F-B-1 sleeving over black and white leads. Solder leads to connector. *Solder black lead to pin "A" and white lead to pin "B". Slide sleeving over connector terminals and wire ends. Use heat gun to shrink in place.

NOTE: Refer to Generator Wiring Schematic Diagram. Some Models "Black &

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- 7) Gently pull excess slack in leads and coil in cavity of spacer. Position connector over spacer, making sure leads are positioned in slot spacer.
- 8) Coat threads of four screws with anti-seize compound and secure screws to housing. Use stainless steel lockwire.

h. Install terminal block as follows:

Position terminal block on stator with stator leads positioned on terminal block studs and install two screws, lock washer and flat washers. Tighten screws. Install washers and nuts on terminal block.

i. Install speed sensor as follows:

Using space connector, connector and screws follow IPL for installation sequence and solder black lead to connector lead B and white lead to connector A.

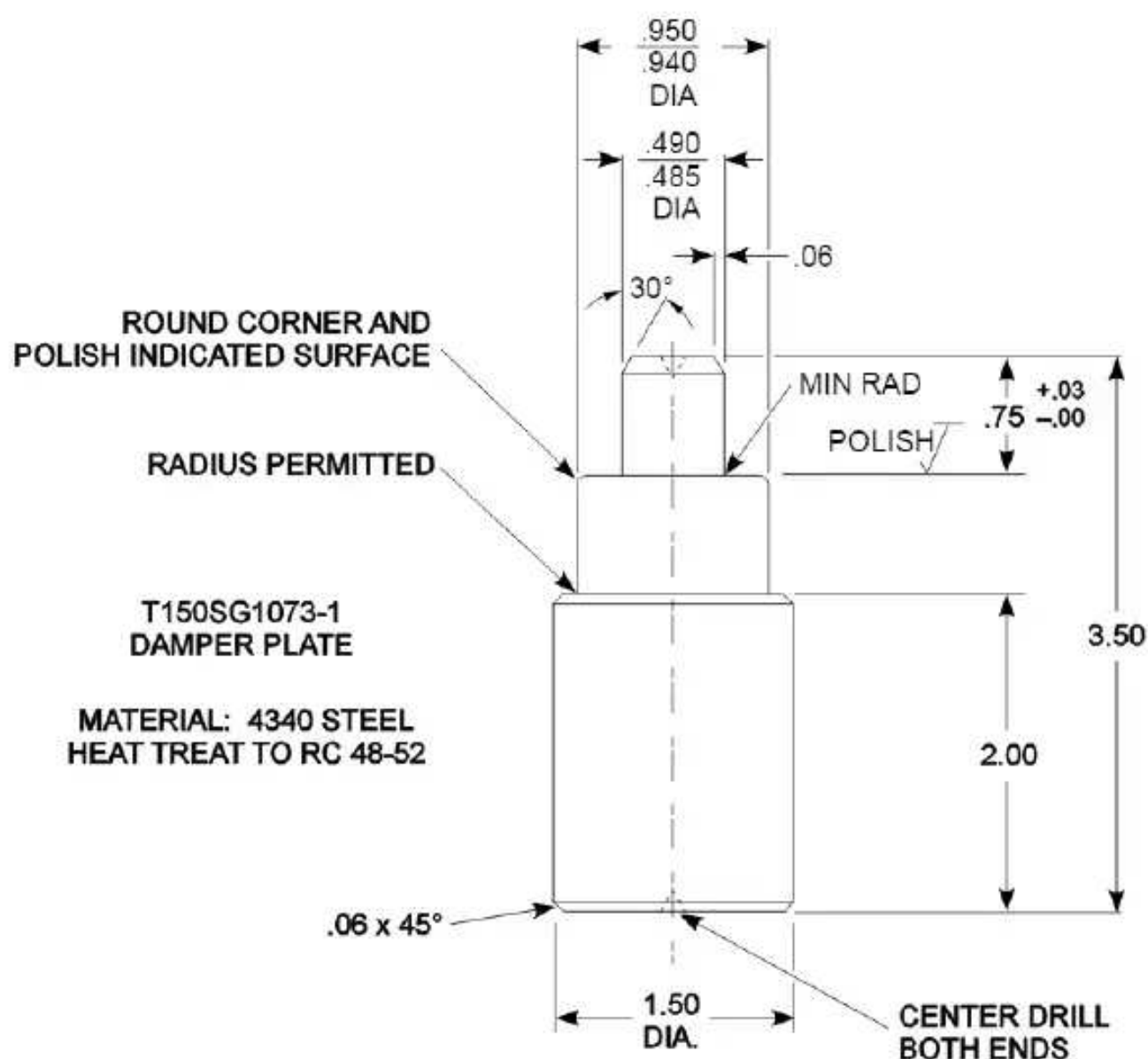
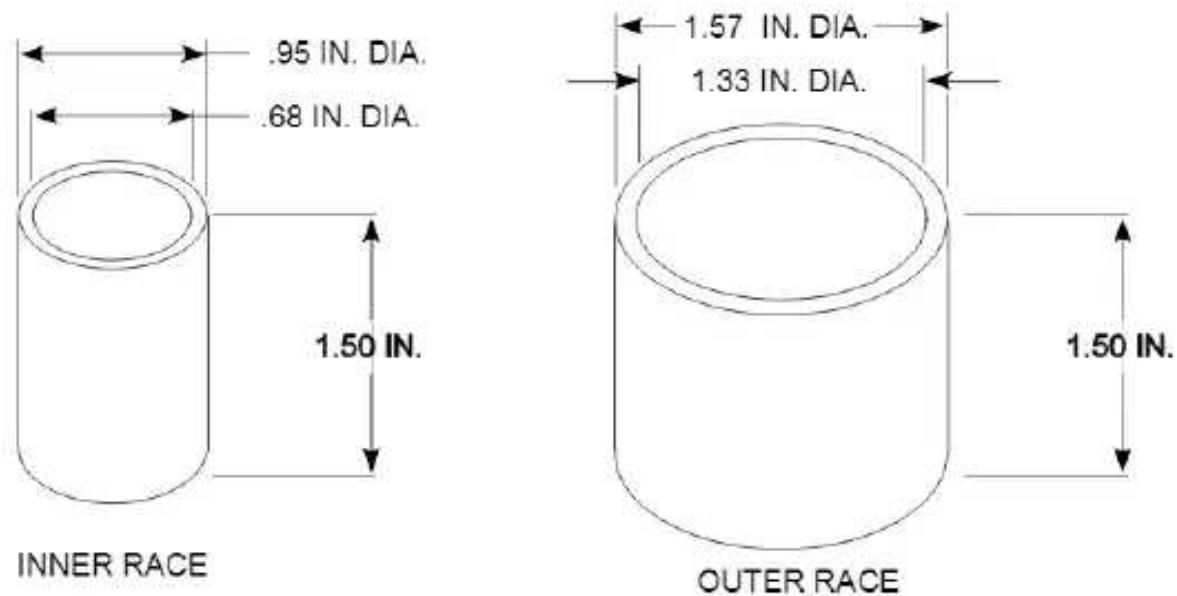


Figure 3-6
Damper Plate Installation Tool (T150SG1073-1)

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MATERIAL: 4340 STEEL
HEAT TREAT TO RC 48-52
TOLERANCE: DECIMALS .XX $\pm$ 0.01

Figure 3-7
Bearing Pusher (Inner Race and Outer Race)

CAUTION: DO NOT HAMMER DAMPER PLATE ON SHAFT OR USE ARBOR PRESS TO FORCE ON THE DAMPER PLATE.

- j. Place damper plate on armature shaft by hand. Make sure damper plate is squarely seated on taper.
- k. Press armature damper plate into position on the armature shaft using a hand arbor press and installation tool T150SG1073-1 (See Figure 3-6). Place the generator drive end up on an aluminum block under the ram of the arbor press. Place tool T150SG1073-1 over the armature drive shaft resting on the damper plate. Using the arbor press, gently press tool T150SG1073-1 until the tool bottoms against the end of the armature shaft and positions the armature damper plate. The damper plate should be square and tight on the armature. If not tight, replace the armature shaft damper plate.
- l. Install four brushes (IPL Figure 5-3) securing brush leads with screws. See Figure 3-11 for correct positioning of brush leads to prevent brush hang-up on the brush box.
- m. Coat the small spline of the drive shaft assembly with Lubriplate 630-AA or equal.
- n. Place a new friction ring on drive shaft. Push shaft into armature making contact with plate on armature shaft.

CAUTION: WHEN BOTTOMING THE NUT, DO NOT OVERTIGHTEN OR DAMPER SPRINGS WILL BE DAMAGED.

- o. Position the fan on the anti-drive end of the drive shaft. Install new Belleville tension springs per Figures 3-8 and 3-9. Reference Data Sheet for quantity. With suitable holding tool, hold drive shaft and install nut and gently tighten until the nut and springs bottom gently against the fan. Then, back off the nut, 3/4 of a turn. Drive shaft should

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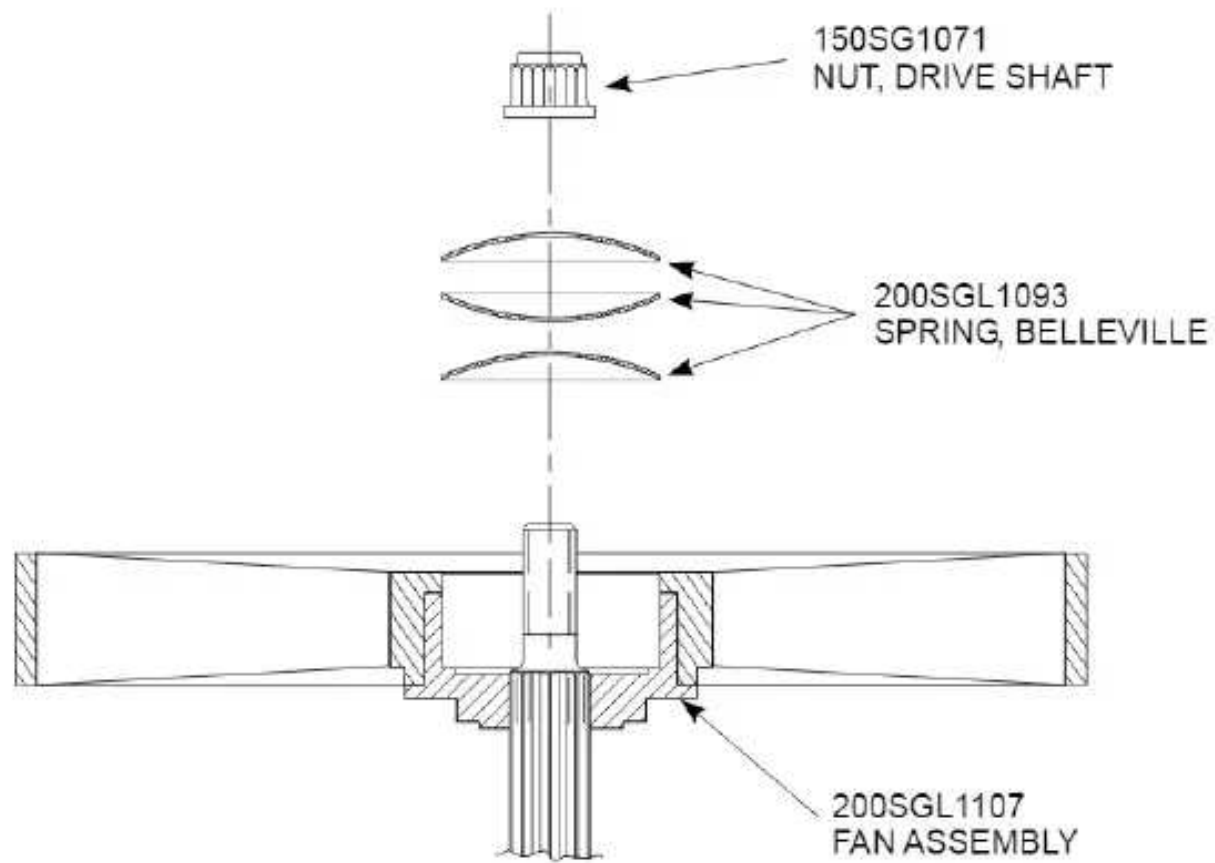


Figure 3-8
Belleville Spring Installation Instructions

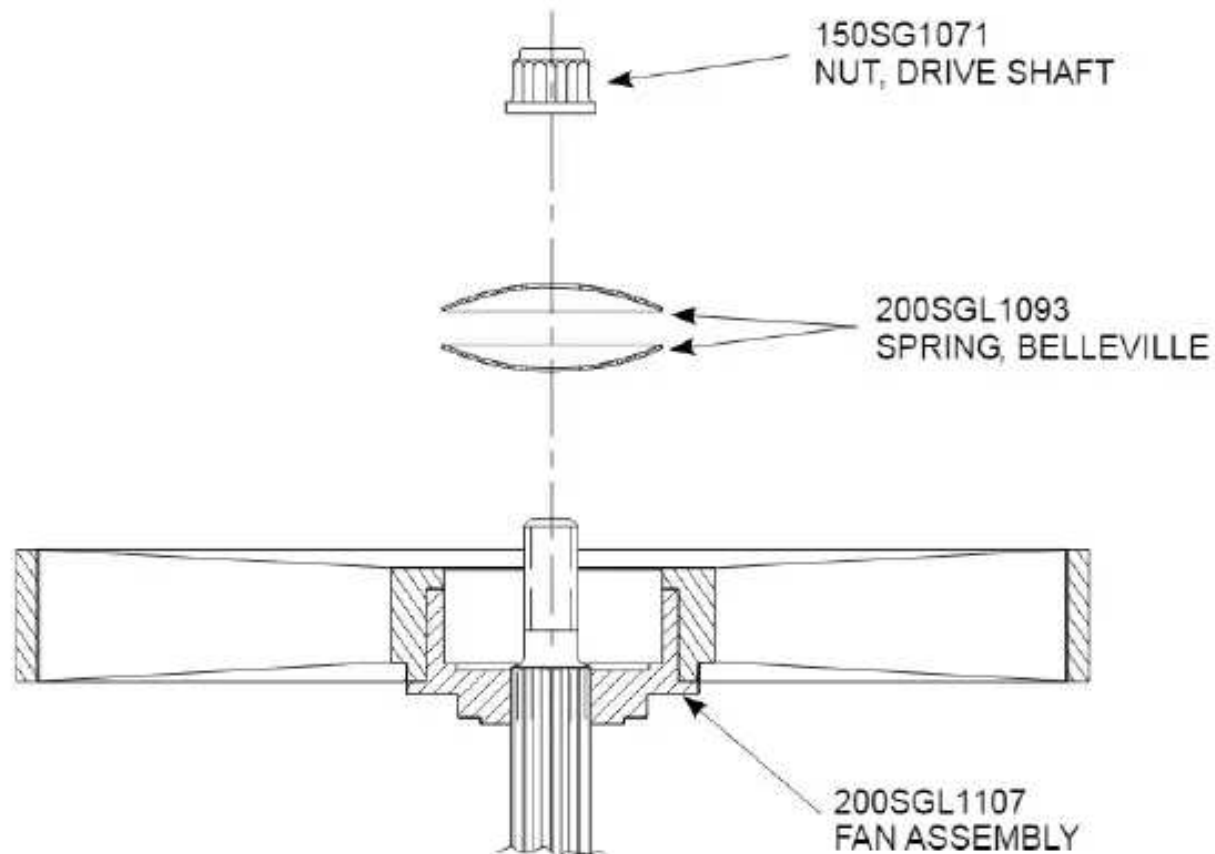


Figure 3-9
Belleville Spring Installation Instructions

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- p. Install fan cover (Figure 5-1) and install four screws.
- q. Install applicable decals per data sheets.

NOTE: The rewinding of the armature or stator in this generator is not recommended.

This armature and stator contain a sophisticated insulation system which is proprietary to Aircraft Parts Corporation.

**ANY REPAIRS OR ALTERATIONS
TO THIS GENERATOR,
NOT COVERED BY THIS MANUAL,
ARE NOT RECOMMENDED.**

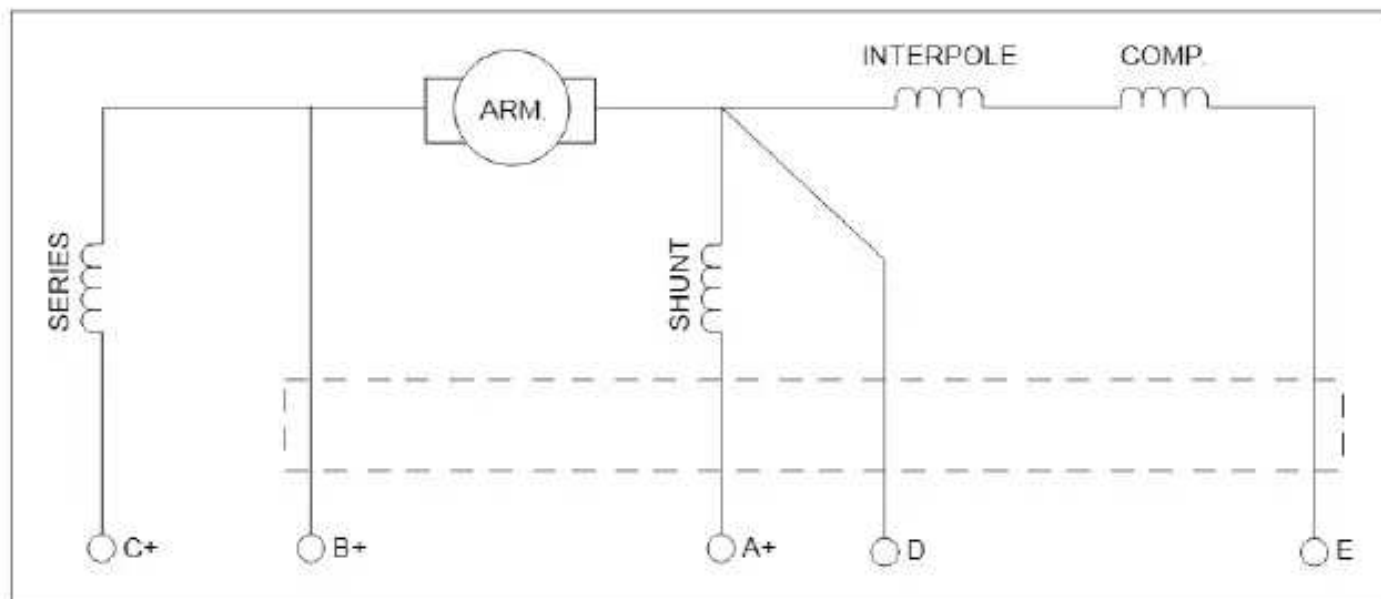


Figure 3-10
Generator Schematic Wiring Diagram

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Figure 3-11
Positioning of Brush Leads

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**GENERATOR SERIES 200SGL
BRUSH LIFE CALCULATION PROCEDURE
BRUSH P/N 200SGL1009**

Date: 16 Dec 1998

Generator P/N: 200SGL117Q

Serial No.: 1200

Calculate brush life left as per example below:

Maximum wear length of a new 150SGL1009-5 Brush (1) .625

Brush wear left (as measured from brush set #1) (2) .310

TOTAL BRUSH WEAR (3) .315

Hours brush has operated: (4) 300

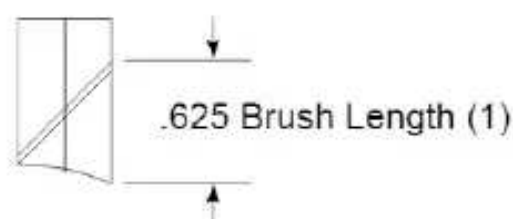
(3) .315 ÷ (4) 300 = (5) .00105 Rate of brush wear per hour
Total Brush Wear (from #3 above) Hours Brush Operated (from #4 above)

(2) .310 ÷ (5) .00105 = (6) 295 Hours of brush left before brushes must be replaced
Brush Wear Left (from Item #2 above) Rate of Brush Wear per Hour (from Item #5 above)

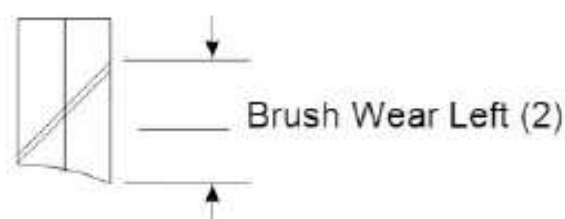
Generator Operating Hours: 295 + 300 = 595
Hours Left (from Item #6) Total Hours on Generator Thus Far (from Item #4) Total Hours Generator Can Operate Before Brush Change

If 250 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above. At that time, return the unit to Unison for overhaul.

NEW BRUSH



BRUSH SET #1



May 12/09

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**GENERATOR SERIES 200SGL
BRUSH LIFE CALCULATION PROCEDURE
BRUSH P/N 200SGL1009**

Date: _____

Generator P/N: _____

Serial No.: _____

Calculate brush life left as per example below:

Maximum wear length of a new 150SGL1009-5 Brush (1) .625

Brush wear left (as measured from brush set #1) (2)

TOTAL BRUSH WEAR (3)

Hours brush has operated: (4)

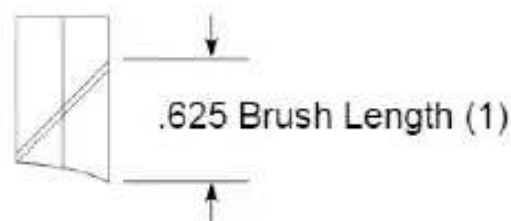
(3) $\div$ (4) = (5) Rate of brush wear per hour
Total Brush Wear (from #3 above) Hours Brush Operated (from #4 above)

(2) $\div$ (5) = (6) Hours of brush left before brushes must be replaced
Brush Wear Left (from Item #2 above) Rate of Brush Wear per Hour (from Item #5 above)

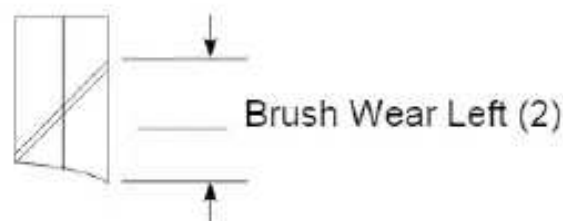
Generator Operating Hours: + =
Hours Left Total Hours on Total Hours Generator
(from Item #6) Generator Thus Can Operate Before
Far (from Item #4) Brush Change

If 250 hours or more of brush life is left, reinstall the brush and operate the generator for the remaining hours of brush wear left, as calculated above. At that time, return the unit to Unison for overhaul.

NEW BRUSH



BRUSH SET #1



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SECTION IV
TESTING AND FAULT ISOLATION

4-1. General

- 4-2. If the unit under test fails to meet any of the following requirements and visual inspection does not disclose the cause of the trouble, disassemble and repeat inspection and testing. After reassembling, repeat procedure from the beginning.

4-3. Test Equipment and Preparation For Test

- 4-4. The standard test equipment required is a generator test stand capable of speeds from 6,500 to 14,000 rpm and a cabinet containing the necessary wiring, switches and meters as shown in Figure 4-1. The load bank should provide graduated load steps and a minimum load rating of 200 amperes.

4-5. Voltage Regulation

For voltage regulation, connect a carbon pile regulator, Generator Control Unit or rheostat capable of controlling the field current from 0 to 10 amperes as shown in Figures 4-1 and 4-2.

4-6. Ambient Temperature

Conduct all tests with an air inlet temperature of 50°F (10°C) to 104°F (40°C). A blower supplying cooling air with a static pressure of 6" inches of water measured in the cooling duct 8" upstream from the generator is required for all tests except minimum speed.

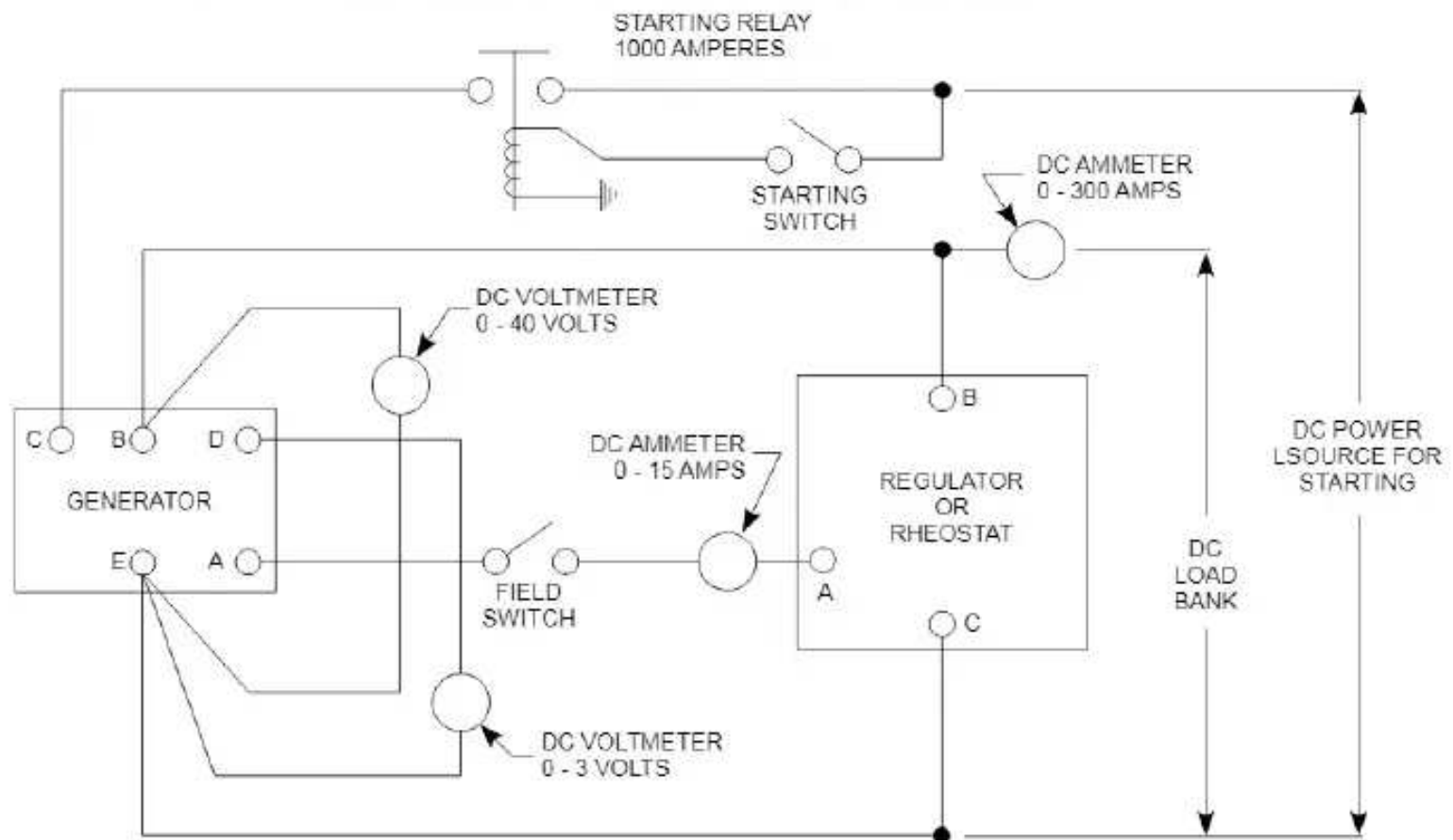


Figure 4-1
Test Schematic For Testing Series Start Generators

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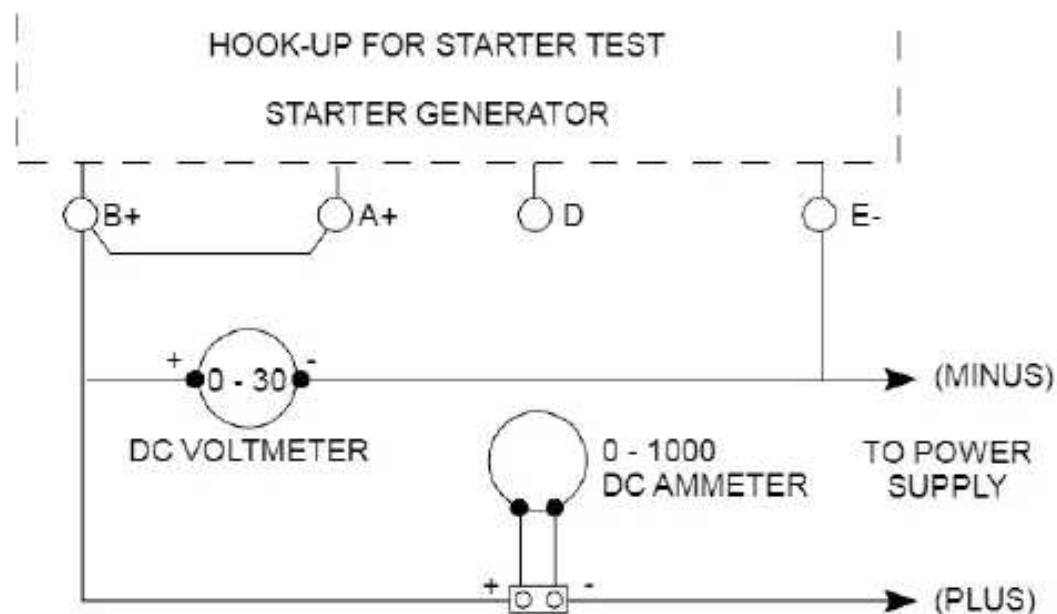
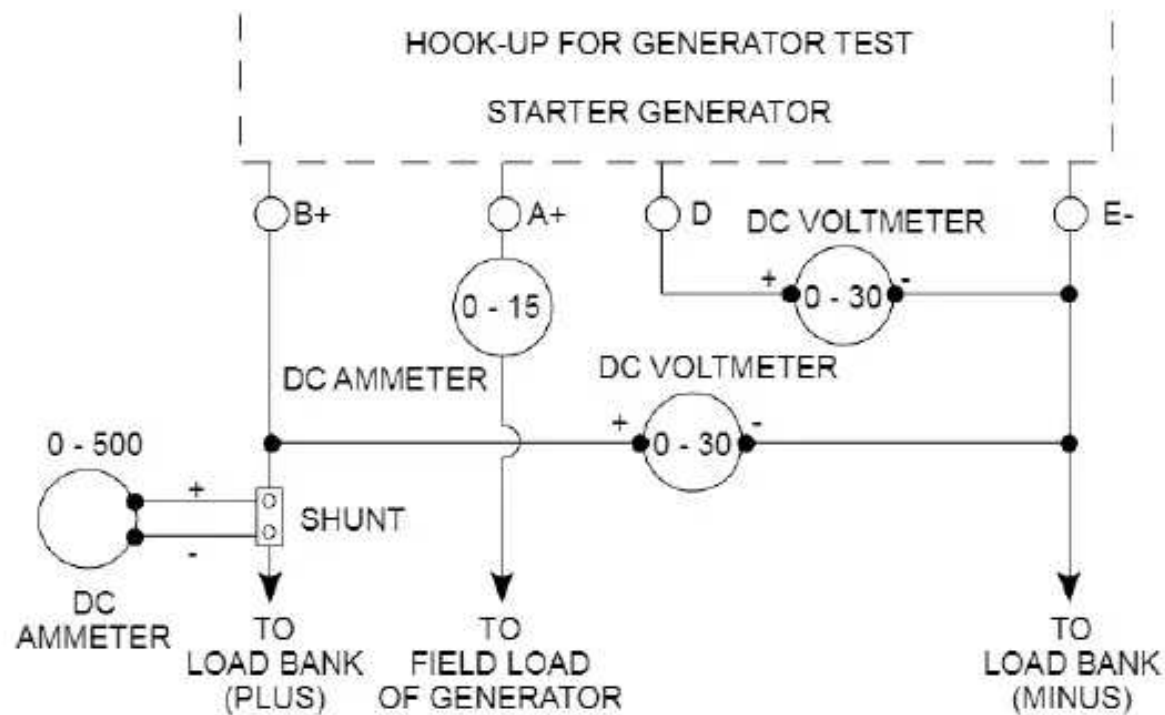


Figure 4-2
Test Schematic For Testing Shunt Start Generators
(without "C" Terminal)

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4-7. Flashing The Field

WARNING: USE A KNIFE SWITCH WHEN FLASHING THE FIELD. OPENING OR CLOSING THE CIRCUIT AT THE GENERATOR OR BATTERY TERMINALS CAN RESULT IN SEVERE DAMAGE TO THE GENERATOR OR BATTERY TERMINALS OR EXPLOSION OF THE BATTERY.

CAUTION: DO NOT FLASH THE FIELD WHILE GENERATOR IS IN OPERATION.

If required, flash the field as follows:

- A. Connect the positive terminal of a 12-volt battery through a single-pole, single-throw knife switch to generator terminal A on the terminal board (Figure 4-1).
- B. Connect the negative terminal of the battery to terminal E on the generator terminal board.
- C. Apply battery current to the field for 5 seconds by closing the knife switch.
- D. Repeat the operation several times to be sure that the field is properly flashed.

4-8. Test Connections

- A. Connect the Starter-Generator to the test circuit, as shown in Figures 4-1 or 4-2 as appropriate.
- B. Move all load switches to the OFF position.
- C. Attach a thermocouple on the outside of the housing, on the side opposite the terminal block, to measure the frame temperature.

4-9. Rotation

- A. Operate the Starter-Generator in a counterclockwise direction, as viewed from the drive-end, during the tests.

4-10. Brush Seating

- A. Brush seating is of extreme importance. If brushes are not seated prior to use, severe arcing will occur during the starting mode of operation causing damage to the commutator surface and accelerated brush wear will occur.
- B. Brushes installed in newly overhauled starter generators should have a commutator resurfaced by turning the armature in a lathe to a very fine mirror finish using a diamond or newly honed carbide turning tool. The best brush life will always be obtained on a resurfaced commutator.
- C. Mount the Starter-Generator securely on a bench. Motorize the starter generator by connecting the positive lead of a DC supply to starter generator terminal C+ (for a series start) or a B+ for a shunt. Connect the negative lead of the DC supply to the starter generator E-. Adjust the DC supply voltage to 15 volts for series wound generators and 30 volts for shunt wound generators. This will enable the generator to run at approximately 7000 – 9000 rpm for a series start, and 3000 – 6000 rpm for a shunt start. The DC amperes

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- D. An acceptable brush seat should indicate 85 – 100% brush contact in the direction in rotation and at least 85% parallel to the armature shaft. Continue running the generator to achieve this seating, if brush seating is unacceptable. Refer to Figure 4-3 for typical brush seating.

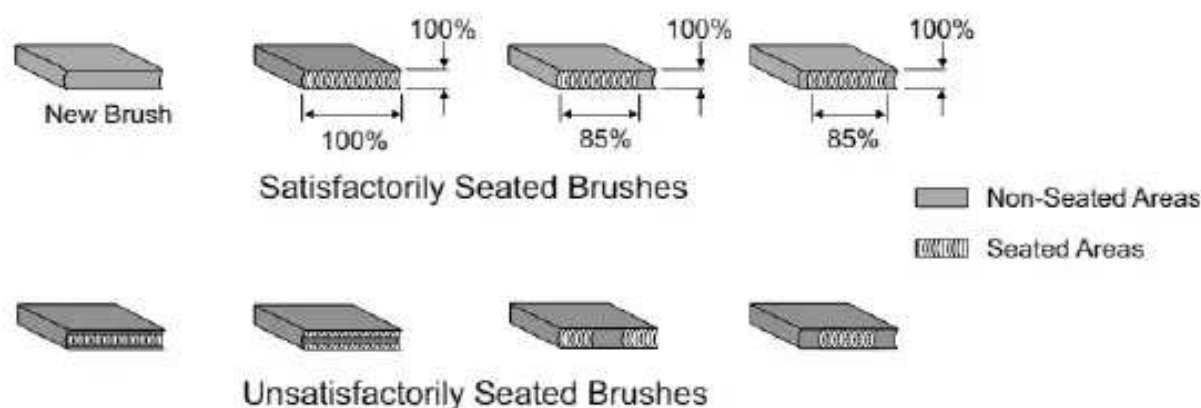


Figure 4-3
Typical Brush Seating

- E. Most APC grades of brushes are instant filming types and during the procedure of brush seating, a dark film will appear on the commutator surface. This film lowers brush friction and increases brush life.
- F. Once the brushes are satisfactorily seated the starter generator is then ready for functional testing.
- G. Brush Seating (Variable Drive Motor)

NOTE: Brush run-in by motorizing the Starter-Generator can be utilized in lieu of Paragraph 4-10.G.

- 1) Start the drive motor and increase the Starter-Generator speed to approximately 8100 rpm as indicated on a tachometer.
- 2) Close suitable switches on the DC load bank to apply 50 amperes load as indicated on the DC load ammeter.
- 3) Operate the Starter-Generator for a minimum of two hours. Shut off the Starter-Generator at the end of this period.
- 4) Examine the brush seating one brush at a time. The face of each brush should contact the commutator surfaces 100 percent in the direction of rotation, and at least 85 percent of the surface parallel to the armature shaft. (See Figure 4-2).
- 5) If the brush seating is close to the required values, increase the DC load between 75 and 125 amperes for short periods while inspecting to see if the brushes are properly seated. See Figure 4-2 for typical brush seating.

NOTE: When using a brush spring lifter when removing the brushes from the brush holders, be sure to return each brush to the brush holder from which it was removed.

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- 6) When the brush seating is completed, open all load switches and shut down the Starter-Generator. Allow the unit to cool before proceeding with tests. Record readings on test data sheet, Table 4-2.

4-11 Test

4-12. Continuous Operating Condition

Continuous operating condition is defined as follows:

While the Starter-Generator is cold, the resistance and temperature of the shunt field shall be determined for use in calculating the average field temperature rise during continuous operation at full load. The Starter-Generator is considered to have reached its continuous operating condition when the rate of rise of the shunt field temperature does not increase by more than 2°F (1.1°C) during a five minute period.

4-13. Maximum Speed For Regulation

The Starter-Generator should not be given an operational warm-up prior to this test. Perform the test as follows:

- A. Connect Starter-Generator to the generator test stand.
- B. Place all load switches to the OFF position.
- C. Increase the speed of the drive motor until a reading of 13,000 rpm $\pm 0/-100$ rpm is obtained on the tachometer. Adjust the field rheostat or carbon pile regulator until generator voltmeter reads 30 ± 0.1 VDC. Shunt field voltage current should not be less than 0.81 amperes minimum.
- D. Reduce the drive motor to zero rpm.
- E. Record readings on test data sheet, Table 4-2.

4-14. Heating, Commutation, Minimum Speed and Equalizing Voltage

- A. Minimum Speed for Regulation (Stabilization Not Required)

Decrease the Starter-Generator speed to 7050 RPM $\pm 100/-0$. The Starter-Generator must deliver 200 amperes at 27-28 volts ± 0.1 . Operate the Starter-Generator (stabilization not required) and record field current, which should not exceed 10.0 amperes. Record readings on test data sheet, Table 4-2.

- B. Continuous Operating Speed

Start the drive motor. Increase the speed of the Starter-Generator to 12,100 RPM ± 100 . Close suitable switches of the DC load bank to apply a load of 200 amperes ± 5 at 30 ± 0.1 volts VDC. When the Starter-Generator reaches its continuous operating conditions as described in Paragraph 4-12, (no more than 2°F [1.1°C] rise in five minutes), check the readings of the temperature instruments. Frame temperature should not exceed air inlet temperature by 175°F (79.44°C). Record readings on test data sheet, Table 4-2.

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C. Equalizing Voltage

Measure the paralleling voltage across terminals D and E (Figure 3-11). See Figures 4-4 and 4-5. Refer to Table 4-1 for which table to use. Record readings on test data sheet, Table 4-2.

Table 4-1
Equalizing Voltage Chart

STARTER GENERATOR	EQUALIZING VOLTAGE CHART
200SGL111Q 200SGL124Q 200SGL111Q-2 200SGL130Q 200SGL114Q 200SGL132Q 200SGL115Q 200SGL144Q 200SGL115Q-2 200SGL145Q 200SGL120Q	USE CHART 1
200SGL111Q-3 200SGL129Q-1 200SGL112Q 200SGL129Q-4 200SGL118Q 200SGL129Q-5 200SGL119Q 200SGL129Q-8 200SGL119Q-2 200SGL142Q 200SGL121Q 200SGL147Q 200SGL129Q	USE CHART 2

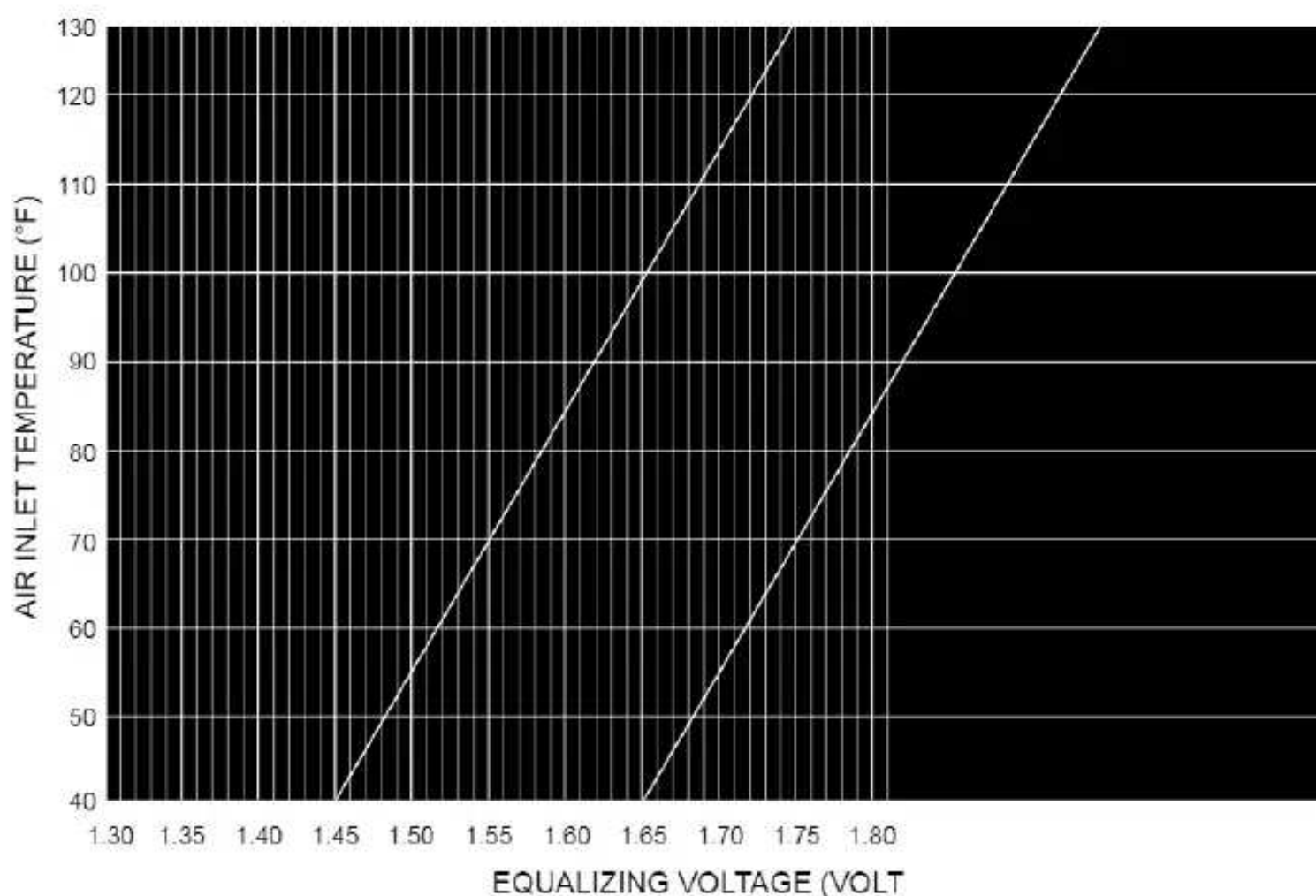


Figure 4-4

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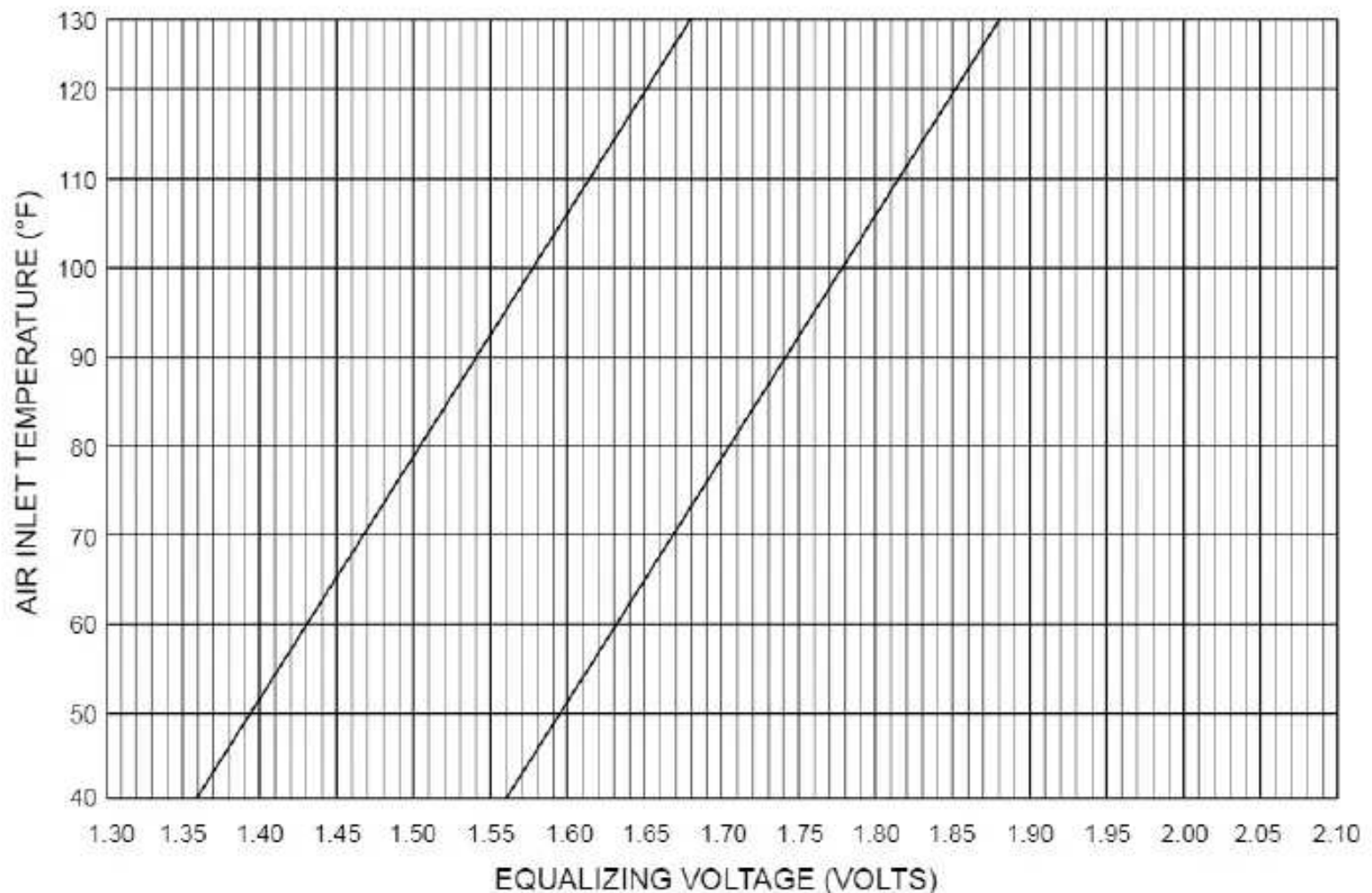


Figure 4-5
Equalizing Voltage (Shunt) Chart 2

4-15. Commutation and Compounding

CAUTION: IF THE PLASTIC WINDOW IS NOT AVAILABLE AND THE TEST IS TO BE PERFORMED WITH THE COVER REMOVED, COMPLETE THE TEST IN 3 TO 4 MINUTES. PROPER COOLING CANNOT BE MAINTAINED WITH THE COVER REMOVED AND THE STARTER-GENERATOR MAY OVERHEAT.

Immediately following the previous heat runs with the Starter-Generator hot, observe the commutation of the Starter-Generator over the speed range of 7050 +100/-0, 10,000 +100/-0 and 12,000 +100/-0 rpm at no-load, half-load, full-load. Sparking must be limited to pin point sparking at the brushes. Commutator should be black for best generator performance. Modify a fan cover to have a plastic window. Rotate the cover to observe each brush when performing this test. If sparking is excessive, check brush seating surfaces and brush tension. The field current must increase with any increase in generator load. Record readings on test data sheet, Table 4-2.

4-16. Speed Pick-up Test

- A. Run the generator at 6,000 rpm with field switch open. Connect a 20 kilohm + 10% load to the speed pick-up output connector between pins B and A and connect an oscilloscope across it. Observe the output voltage.
- B. Acceptance Limits: The peak to peak voltage shall be (2.5) volts minimum. Typical acceptable voltage wave form shall be shown in Figure 4-6.

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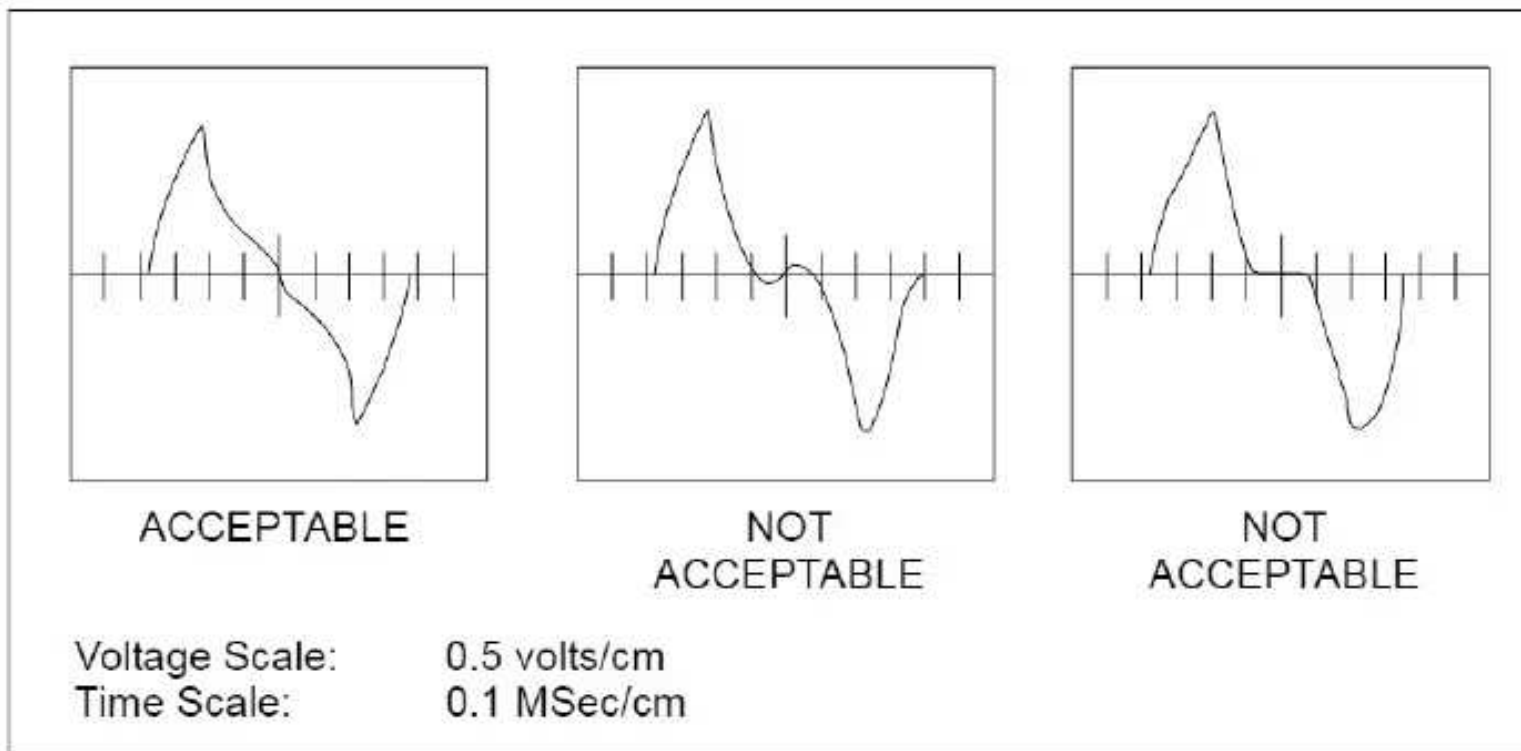


Figure 4-6
Typical Speed Pick-Up Output Voltage Waveform

4-17. Overspeed

- A. Perform this test with the Starter-Generator hot as a result of testing.
- B. Open all load switches and the field switch.
- C. Increase the speed of the Starter-Generator to 14,000 rpm. The Starter-Generator must operate at this speed for 5 minutes without mechanical failure, throwing of varnish or impairment of electrical performance.
- D. At the end of 5 minutes overspeed run, reduce the speed to 12,000 rpm. Close suitable load switches to obtain rated load of 200 amperes.
- E. Operate the Starter-Generator for 1 minute and check the electrical performance which should compare with that observed in paragraph 4-15. Record readings on test data sheet, Table 4-2.

4-18. Dielectric Strength Test (IPL Fig. 5-1)

NOTE: If the generator test stand capable of 14,000 RPM is not available, the overspeed test can be performed by energizing the starter generator as a motor and adjusting the no load speed to the required 14,000 RPM.

Perform this test while the Starter-Generator is hot as a result of testing.

CAUTION: DO NOT APPLY VOLTAGE ACROSS SHUNT, SERIES INTERPOLE AND COMPENSATING FIELDS AS DAMAGE MAY OCCUR.

- A. Remove all external connections from the terminals. Disconnect the terminal block by

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200SGL SERIES STARTER GENERATOR

- B. Using a Hi-Pot, apply 250 VAC, 60 Hz between E and the stator housing (ground).
- C. Apply the same voltage between series field and interpole and compensating field terminals (D, E, B and C) and the stator housing (ground).
- D. The insulation should not breakdown.
- E. After the dielectric strength test, install and reconnect the terminal block. Reconnect wire terminals and install two screws.
- F. Record readings on test data sheet, Table 4-2.

4-19. Acceleration Run

- A. Connect to an acceleration test stand with a steel flywheel 12 inches in diameter and 3.187 inches thick. A 0.836 inch flywheel may be used in lieu of 3.187 inch thick flywheel. The flywheel should rotate on ball bearings sized to the weight and RPM of the wheel.
- B. The Starter-Generator must accelerate the flywheel to the indicated speed within allotted time as listed on the acceptance test data sheet. Do not exceed 1000 amps max (for 200SGL Series).
- C. Record peak in rush current (1000 amps maximum) and time to reach allowed rpm on test data sheet. Record readings on test data sheet, Table 4-2.

4-20. Commutator Film

It is important for brush life and good generator operating characteristics that the commutator have a black film. Inspect the commutator after the final test for a black film. If the film is not evident, run the generator at 10,000 rpm and 200 amperes for 30 minutes.

4-21. Commutator Run-Out

Using a dial indicator firmly mounted to the generator and with the probe located on the commutator, the total commutator run-out shall be .0008 in. (.0203 mm) maximum. Bar to bar run-out shall be .0002 in. (.0050 mm) maximum. Record readings on test data sheet, Table 4-2.

4-22. Vibration

Radial Vibration. Place generator on suitable stand, accelerate to 4000-4500 RPM no load speed and check with Metrix miniature vibration meter or equivalent just below terminal block. Vibration shall not exceed 0.001 in. (.0025 mm) total deflection as a starter. It shall be permissible to employ a drive shaft support during the test to keep the drive shaft in axial alignment. Record readings on test data sheet, Table 4-2.

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Table 4-2
200SGL Series Acceptance Test Data Sheet

REPAIR STATION NAME _____

ADDRESS _____

TESTED PER TM105A

REV _____

REV DATE _____

REPAIR STATION CERTIFICATE NO. _____

S/G MODEL NO. 200SGL _____	SERIAL NO. _____	SALES/WORK ORDER NO. _____	TESTED BY _____	DATE: TEST STAND NO: _____
MOD / REV _____				

TEST PARA.	INSPECTION / TEST	REQUISITES	MEASUREMENT	ACC	REJ
4-10	Brush Seating	100% Seating all Brushes (See Fig. 4-3)	X	X	
4-13	Maximum Speed for Regulation 13,000 +0/-100 RPM No Load, 30.0 ± 0.1 VDC	Ammeter readout (A+) 0.81 amps min.		amps	
4-14A	Minimum Speed for Regulation 7,050 +100/-0 RPM 27-28 ± 0.1 VDC, 200 ± 5 Amp	Field Current (A+ within 10.0 amps)		amps	
4-14B,C	Continuous Operating Speed and Equalizing Voltage 12,100 ± 100 RPM, 200 ± 5 Amp, 30 ± 0.1 VDC	D - E Voltage (See Figs 4-4 and 4-5)		vdc	
		Stabilized Temp. (gen. yoke)		°F	
		Air In Temp.		°F	
4-15	Commutation 27 VDC min. and Compounding	7050 ± 100/-0 RPM, 0, 100, 200 (± 5) Amp	Not to exceed pinpoints, Field Current rises with Load	X	X
		10,000 ± 100 RPM, 0, 100, 200 (± 5) Amp	Not to exceed pinpoints, Field Current rises with Load	X	X
		12,000 ± 100 RPM, 0, 100, 200 (± 5) Amp	Not to exceed pinpoints, Field Current rises with Load	X	X
4-16	*Speed Pickup (6000 ± 100 RPM)	Output 2.5 volts peak to peak min.	X	vdc	
		Wave form shape	X	X	
4-17	Overspeed 5 min 14,000 ± 100 RPM	No Sign of Failure	X	X	
	Post Overspeed 12,000 ± 100 RPM, 30 ± 0.1 VDC, 200 ± 5 Amps	No Sign of Failure	X	X	
4-18	Dielectric Strength 250 ± 10 VAC, RMS, 1.0 sec	No Sign of Failure	X	X	
4-19	Acceleration Run (cranking) 28.5 + 0/-5 VDC Supply Voltage	Peak Current (1000 amps max)		amps	
		0 - 6500 ± 100 RPM within 36 seconds for series starter generators.		sec	
		0 - 6500 ± 100 RPM within 50 seconds for series starter generators, with optional 3.187 inch thick flywheel.		sec	
		0 - 3800 ± 100 RPM within 30 seconds for shunt starter generators.			
		0 - 3800 ± 100 RPM within 40 seconds for shunt starter generators, with optional .836 thick Flywheel.			
4-21	Commutator Runout	Bar-to-Bar (not to exceed 0.0002 inch)		inch	
		TIR (not to exceed 0.0005 inch)		inch	
4-22	Radial Vibration, 4000 - 4500 RPM (when applicable)	Within Limits (0.001 inch total max)		inch	

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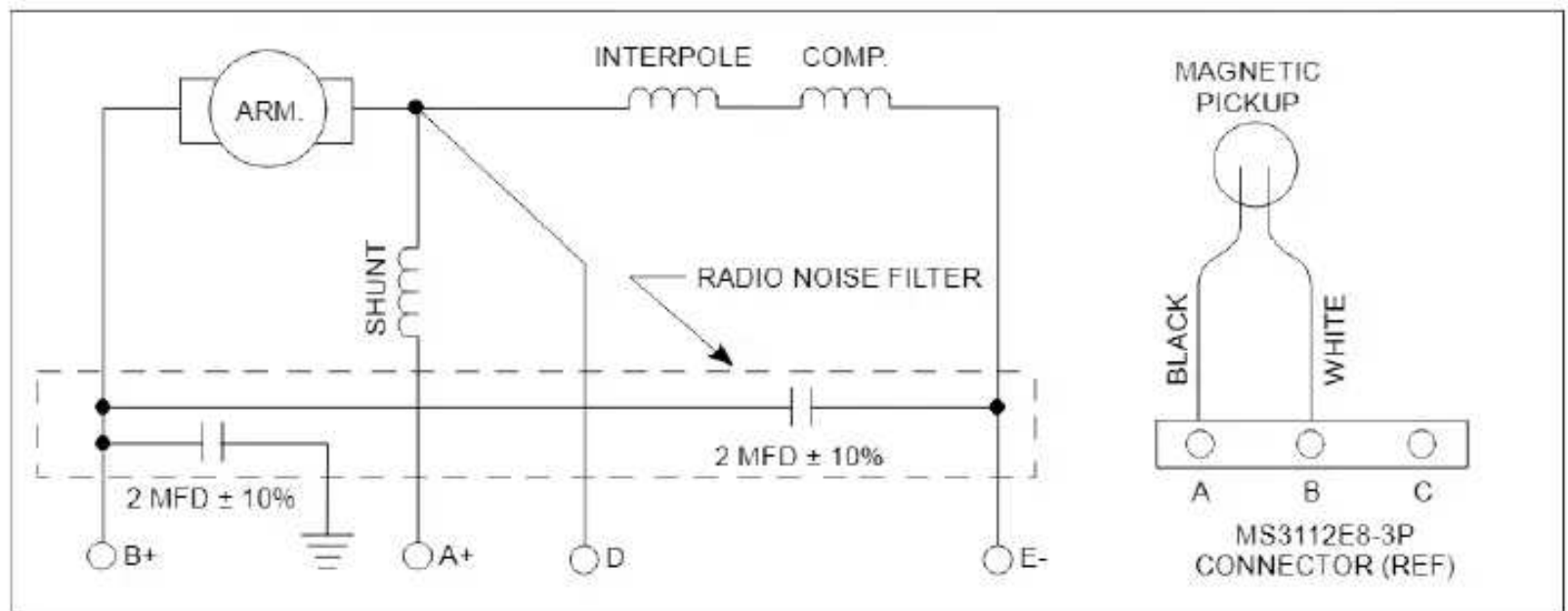


Figure 4-7
Terminal Block & Generator Schematic Wiring Diagram

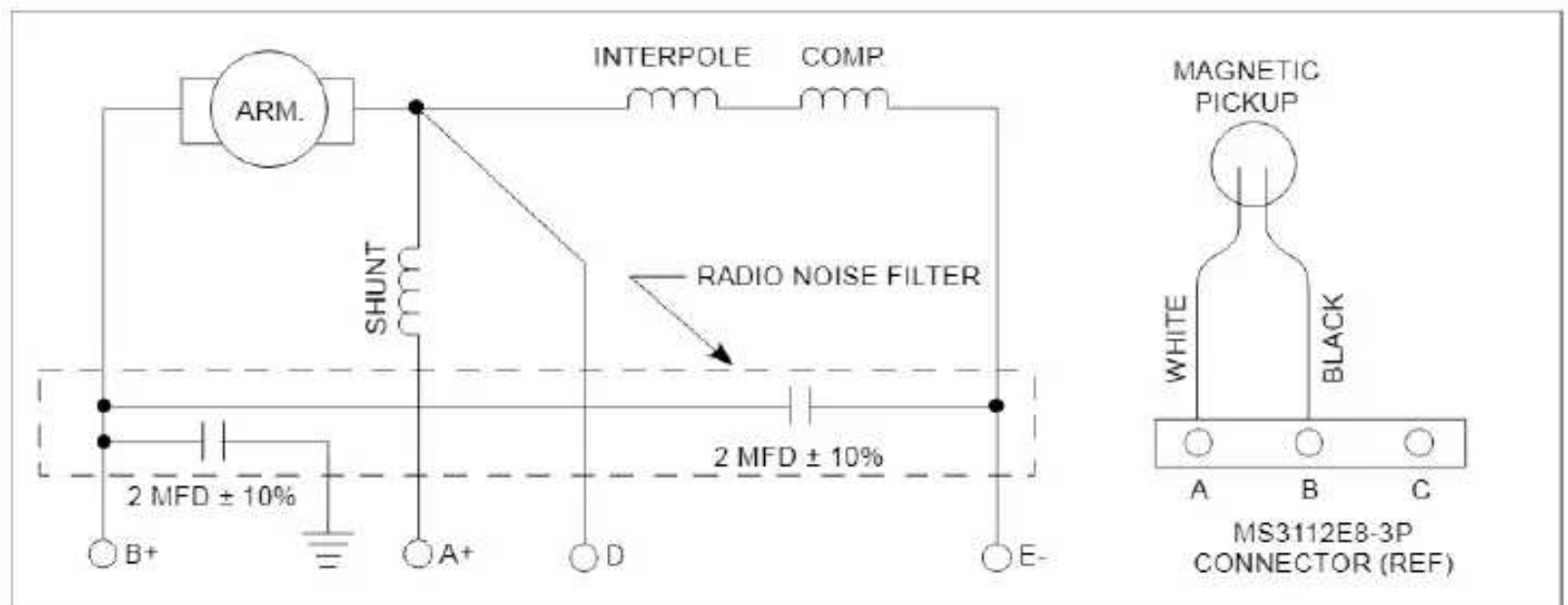


Figure 4-8
Terminal Block & Generator Schematic Wiring Diagram

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OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

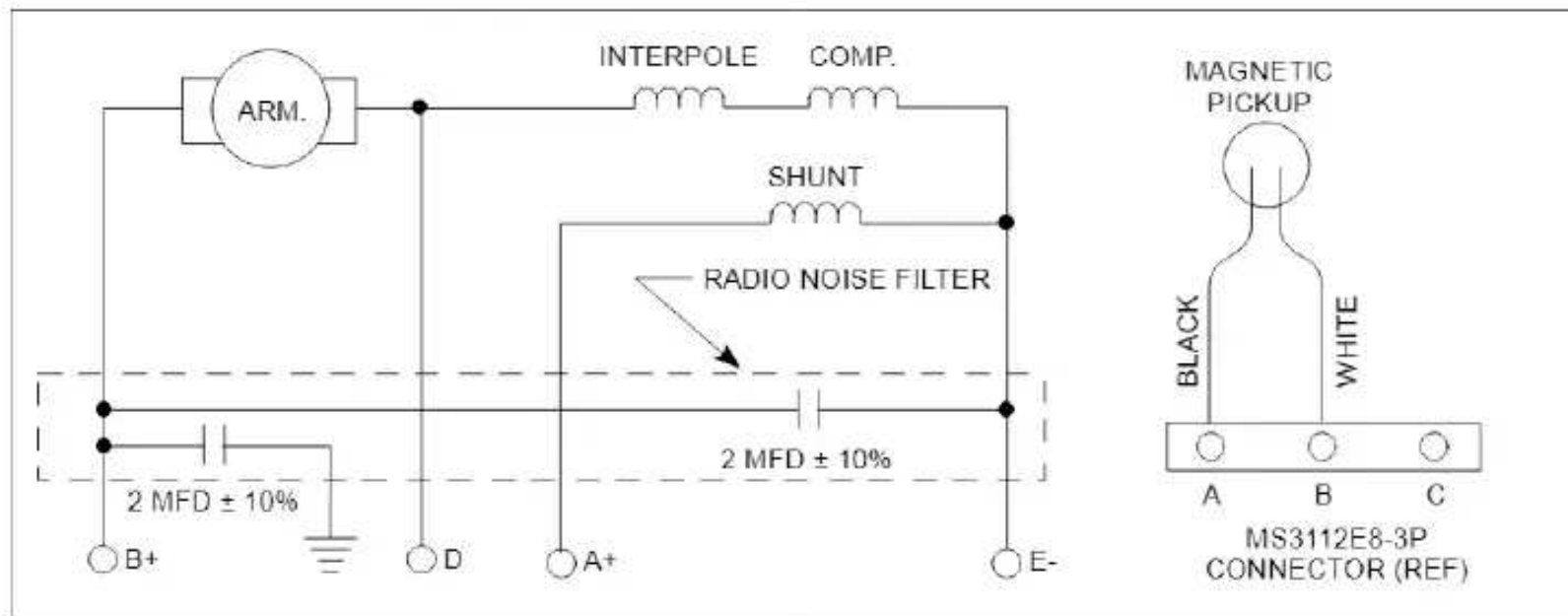


Figure 4-9
Terminal Block & Generator Schematic Wiring Diagram

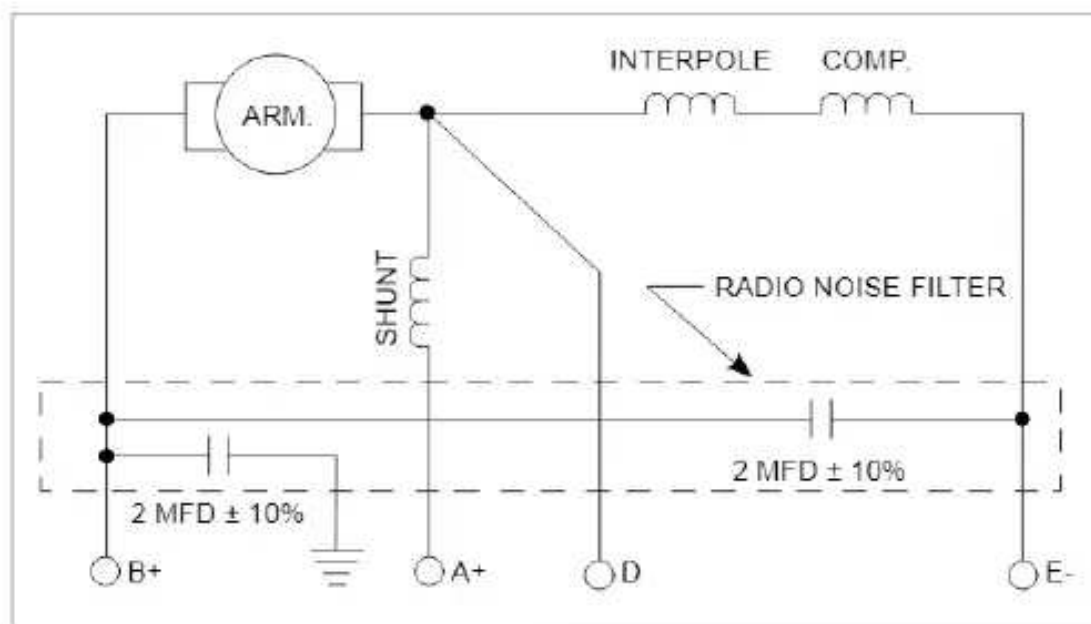


Figure 4-10
Terminal Block & Generator Schematic Wiring Diagram

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200SGL SERIES STARTER GENERATOR

SECTION V
ILLUSTRATED PARTS CATALOG

5-1 General

- A. These drawings are to be used in conjunction with the individual data sheets for parts referencing.
- B. IPL Number (found on data sheets)

The IPL numbers and drawings indicate the figure on which the part is illustrated and its identification on the illustration.

- C. Part Numbers

This column contains numbers for identification of parts.

- D. Description

Description of each part number.

- E. Units Per Assembly

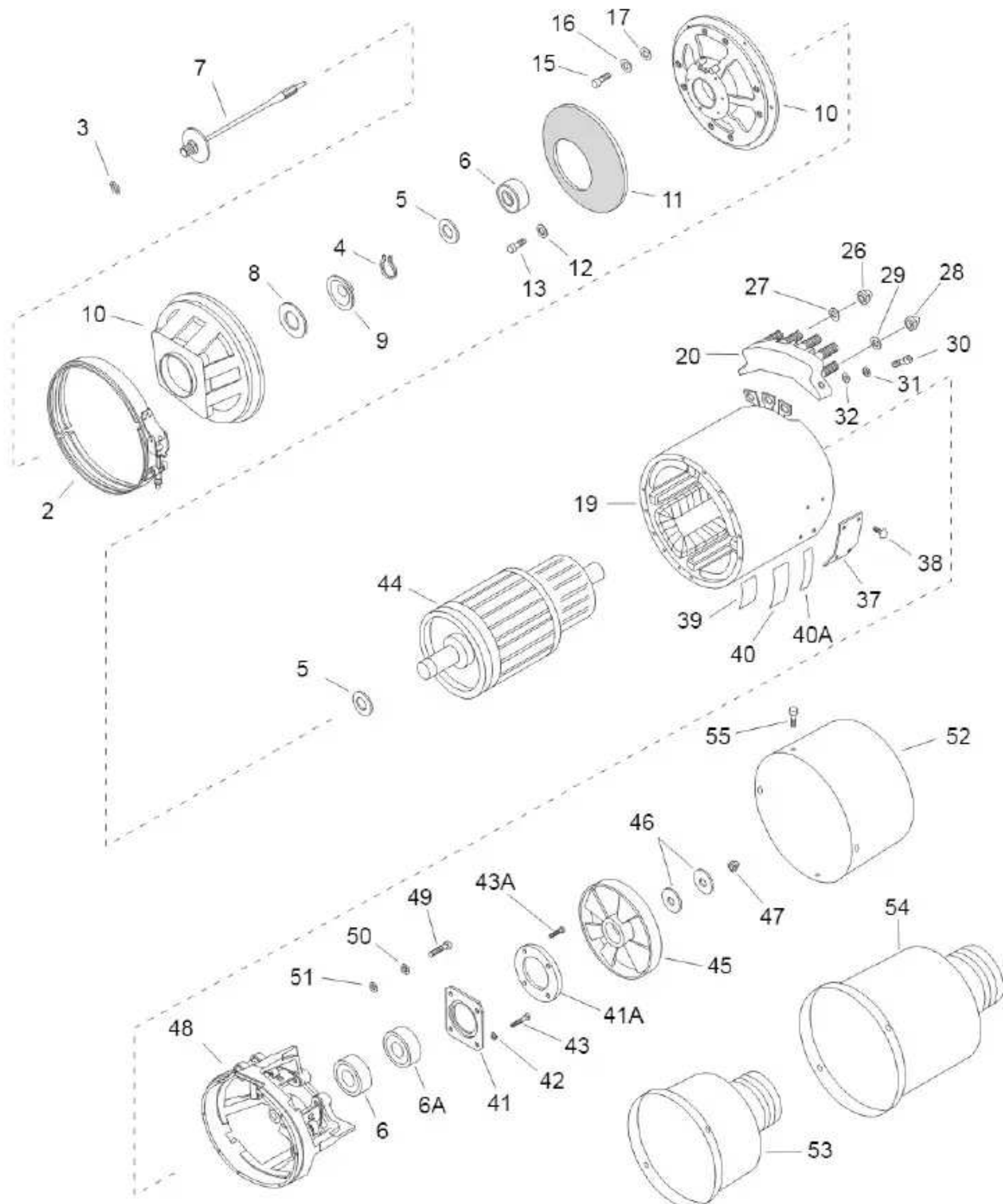
The numbers listed in the "Quantity" column indicates the quantity of parts used per assembly at the location shown.

- F. Federal Supply Code to Manufacturers listing:

The following are vendor approved CAGE code numbers and addresses for vendors listed in the detailed parts Lists:

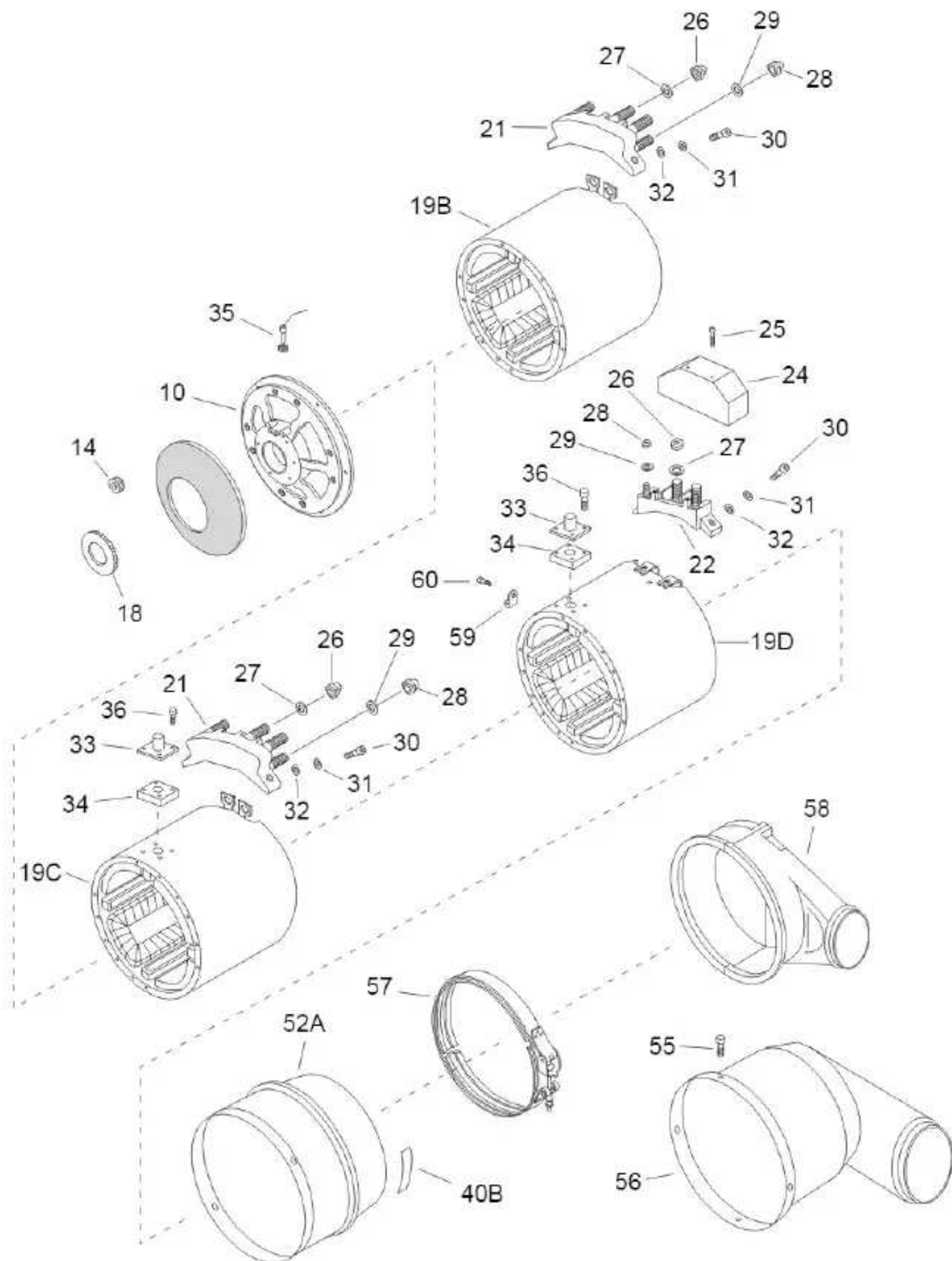
<u>CODE</u>	<u>MANUFACTURER</u>
56878	SPS Technologies 301 Highland Avenue Jenkintown PA 19046

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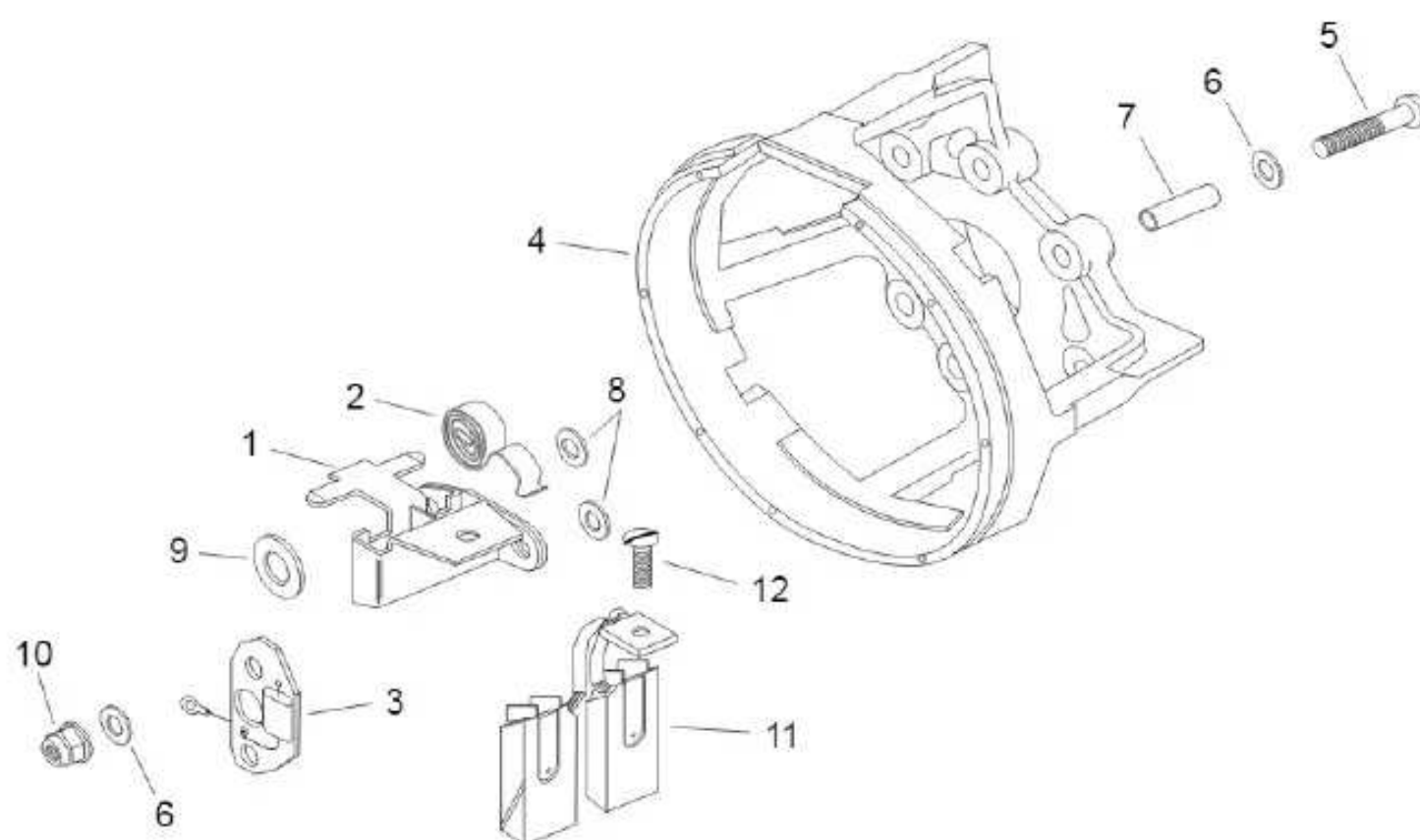
Starter Generator

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200SGL SERIES STARTER GENERATOR



Starter Generator - Model Differences
Figure 5-2

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200SGL SERIES STARTER GENERATOR



Starter Generator - Bell Assembly, Commutator End
Figure 5-3

AIRCRAFT PARTS CORP.
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200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL111Q

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL111Q	Series	200SGL1083	Wet	28-Jul-81

Installations

Type Certificates

MD Helicopter- 530 Series
Bell Helicopter- 206L-1
Rocket Engineering- Piper Turbo Conversion

Supplemental Type Certificates

SH123NE Bell 206L-1 10 February, 1983
 (Amended: 3 May, 1993)
SH401NE Bell 206L-3 19 November, 1985
 (Amended: 24 March, 1993)

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance:

1.80 - 2.06 Ohms

PC Board Capacitance tolerance:

.33 MFD +/- 10%

Terminal block Capacitance tolerance:

B - E -1.8 to -2.2 UFD
B - C -.015 to -.0238 UFD
E - Grnd -.015 to -.0235 UFD
B-Grnd. -.016 to -.0240 UFD
C-Grnd. -1.8 to -2.2 UFD
C - E -.015 to -.0236 UFD

Hi-Pot Test (Stator)

250 VAC for 1 second
Max leakage current allowable: 2.5 milliamps
A-Grnd., A-B, A-E, A-C
B-Grnd., B-E
C-Grnd.
E-Grnd.

Equalizing Voltage Chart

Chart 2

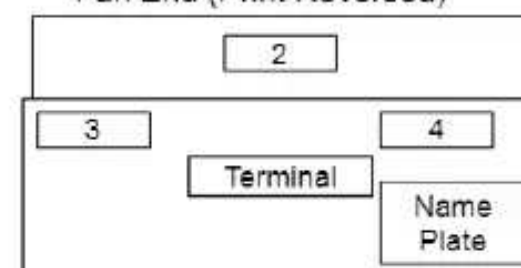
Modified Status

Applicable Decals

Decals 3

- | | |
|-------------|--|
| 2
3
4 | Hi-Pot Warning
APC XL2 Label
QAD Warning |
|-------------|--|

Fan End (Print Reversed)



Other Applicable Technical Data:

SB150SG128-1

XL2 Brushes

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL111Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL111Q		STARTER GENERATOR		RF
5-1-	200SGL1083		. QAD KIT		1
5-1-1	200SGL1040		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-B		. BALL BEARINGS		2
5-1-7	200SGL1103		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-1		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-1		. . DRIVE END BELL		1
5-1-11	150SG1005-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		4
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-	200SGL1126		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156		. . STATOR ASSY		1
5-1-20	200SGL1061		. . AXIAL TERMINAL BLOCK		1
5-1-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		3
5-1-27	AN960C616		. . . TERM. BLK WASHER		3

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL111Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-1-29	AN960C10		. . . TERM. BLK WASHER		2
5-1-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-1-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-1-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1092		. . NAME PLATE		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	MS35265-45		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064		. ARMATURE ASSY		1
5-1-45	200SGL1107		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-52	200SGL1014		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL111Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10XL2		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL115Q-2

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.

OVERHAUL MANUAL WITH IPL

200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL115Q-2	Series	200SGL1083	Wet	10-Feb-83

Installations

Type Certificates

MD Helicopters- MD 500, 530, 600N

Supplemental Type Certificates

SH131NE

Sikorsky S-76A, S-76C 26 April, 1983
(Amended 1 September, 1994)

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance:

1.80 - 2.06 Ohms

PC Board Capacitance tolerance:

.33 MFD +/- 10%

Terminal block Capacitance tolerance:

B - E -1.8 to -2.2 UFD

B - C -.015 to -.0238 UFD

E - Grnd. -.015 to -.0235 UFD

B-Grnd. -.016 to -.0240 UFD

C-Grnd. -1.8 to -2.2 UFD

C - E -.015 to -.0236 UFD

Hi-Pot Test (Stator)

250 VAC for 1 second

Max leakage current allowable: 2.5 milliamps

A-Grnd., A-B, A-E, A-C

B-Grnd., B-E

C-Grnd.

E-Grnd.

Equalizing Voltage Chart

Chart 1

Modified Status

Applicable Decals

Decals 3

Fan End (Print Reversed)

2

Hi-Pot Warning

3

APC XL2 Label

4

QAD Warning

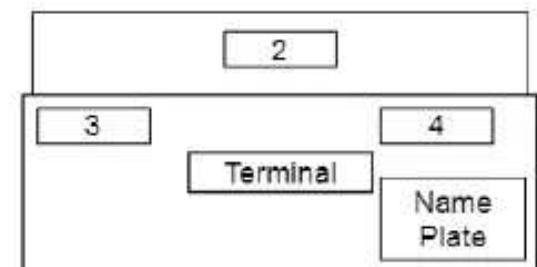
Other Applicable Technical Data:

SB150SG128-1

XL2 Brushes

MD Helicopters Name Plate P/N 200SGL1092-15

Government Name Plate P/N 200SGL1092-9



AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL115Q-2

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL115Q-2		STARTER GENERATOR		RF
5-1-	200SGL1083		. QAD KIT		1
5-1-1	200SGL1040		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-B		. BALL BEARINGS		2
5-1-7	200SGL1103		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-1		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-1		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		4
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-	200SGL1126		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156		. . STATOR ASSY		1
5-1-20	200SGL1061		. . AXIAL TERMINAL BLOCK		1
5-1-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		3
5-1-27	AN960C616		. . . TERM. BLK WASHER		3

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL115Q-2 (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-1-29	AN960C10		. . . TERM. BLK WASHER		2
5-1-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-1-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-1-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1092-3		. . NAME PLATE		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	MS35265-45		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064		. ARMATURE ASSY		1
5-1-45	200SGL1107		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-54	200SGL1014		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL115Q-2 (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10XL2		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL119Q-2

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.

OVERHAUL MANUAL WITH IPL

200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL119Q-2	Shunt	200SGL1083-2	Wet	28-Oct-07

Installations

Type Certificates

Cessna

208 Caravan

Supplemental Type Certificates

SH1039NE

Bell- 212, 412 19 January, 1993
(Amended: 1 December, 2003 Bell
412EP/412CF)

Disassembly and Assembly

Speed Sensor hookup:

Uses magnetic pickup and spur gear

Black lead = A White lead = B (Reverse for Cessna aircraft)

Testing

Shunt field terminal A - D resistance tolerance:

1.80 - 2.06 Ohms

PC Board Capacitance tolerance:

.33 MFD +/- 10%

Terminal block Capacitance tolerance:

+B to -E

E- to Grnd.

B+ to Grnd.

Hi-Pot Test (Stator)

300 VAC for 1 second

Max leakage current allowable: 2.5 milliamps

A-Grnd., A-B, A-E

B-Grnd., B-E

E-Grnd.

Equalizing Voltage Chart

Chart 2

Modified Status

Mod status = XXXX

Description

Date

Change

Stamp Block

SB150SG116

7-Jun-94

PU Leads

#1

19-SB150SG117 Rev. B

30-Dec-02

Pick up

#2

19-SB150SG129

4-Oct-01

Damper

#3

SB200SG141

Washers &

#4

XL2 Brushes

Applicable Decals

Decals 4

Fan End (Print Reversed)

1

Mod Status

2

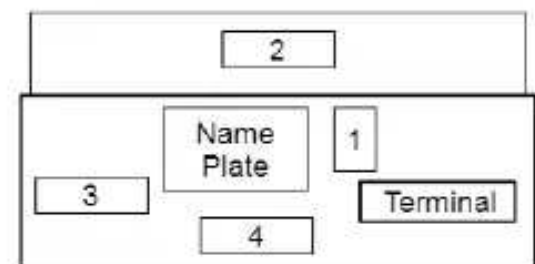
Hi-Pot Warning

3

APC XL2 Label

4

QAD Warning



Other Applicable Technical Data:

Change Armature P/N to XL Series as needed

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL119Q-2

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL119Q-2		STARTER GENERATOR		RF
5-1-	200SGL1083-2		. QAD KIT		1
5-1-1	200SGL1040-2		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-B		. BALL BEARINGS		2
5-1-7	200SGL1103-3		. DRIVE SHAFT ASSY		1
5-1-8	200SGL1070		. FRICTION RING		1
5-1-9	200SGL1073		. DAMPENER PLATE		1
5-1-	200SGL1301-2		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-2		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		6
5-2-14	MS35489-1		. . GROMMET		1
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-18	200SGL1219-1		. SPUR GEAR		1
5-2-	200SGL1126-1		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156-5		. . STATOR ASSY		1
5-2-22	200SGL1061-2		. . RADIAL TERMINAL BLOCK		1

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL119Q-2 (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-24	200SGL1318		. . COVER TERMINAL BLOCK		1
5-2-25	NAS1189-06P12B		. . TERM. BLK COVER SCREW		2
5-2-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		2
5-2-27	AN960C616		. . . TERM. BLK WASHER		2
5-2-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-2-29	AN960C10		. . . TERM. BLK WASHER		2
5-2-30	MS35265-66		. . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-2-33	MS3112E8-3P		. . CONNECTOR		1
5-2-34	200SG1221		. . SPACER CONNECTOR		1
5-2-35	300SG1222-2		. . MAGNETIC PICKUP		1
5-2-36	MS35265-19		. . CONNECTOR SCREW		4
5-1-37	200SGL1092-51		. . NAME PLATE		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-43	AN500A8-8		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064-1-XL		. ARMATURE ASSY		1
5-1-45	200SGL1107		. FAN ASSY		1
5-1-46	250SGL1093		. BELLEVILLE SPRINGS		3
5-1-47	150SG1071		. SELF LOCKING NUT		1

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL119Q-2 (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-54	200SGL1011		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-2-59	400SG1237		. CABLE CLAMP		1
5-2-60	MS35266-59		. CABLE CLAMP SCREW		1
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-4		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL124Q

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.

OVERHAUL MANUAL WITH IPL

200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL124Q	Series	200SGL1083	Wet	24-Jul-86

Installations

Type Certificates

U.S. Army
Bell Helicopter

Supplemental Type Certificates

None

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance:

1.80 - 2.06 Ohms

PC Board Capacitance tolerance:

.33 MFD +/- 10%

Terminal block Capacitance tolerance:

B - E
B - C
E - Grnd.
B-Grnd.
C-Grnd.
C - E

Hi-Pot Test (Stator)

250 VAC for 1 second
Max leakage current allowable: 2.5 milliamps
A-Grnd., A-B, A-E, A-C
B-Grnd., B-E
C-Grnd.
E-Grnd.

Equalizing Voltage Chart

Chart 1

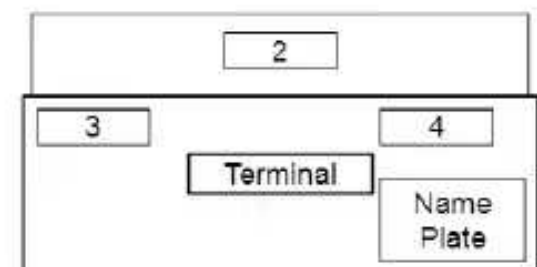
Modified Status

Applicable Decals

Decals 3

Fan End (Print Reversed)

- 2 Hi-Pot Warning
- 3 APC XL2 Label
- 4 QAD Warning



Other Applicable Technical Data:

SB150SG128-1

1,000 Hr TBO with 500 Hr Mid-life brush change

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL124Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL124Q		STARTER GENERATOR		RF
5-1-	200SGL1083		. QAD KIT		1
5-1-1	200SGL1040		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-B		. BALL BEARINGS		2
5-1-7	200SGL1103-11		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-1		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-1		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		6
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-	200SGL1126-6		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156		. . STATOR ASSY		1
5-2-22	200SGL1061-4		. . AXIAL TERMINAL BLOCK		1
5-2-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		3
5-2-27	AN960C616		. . . TERM. BLK WASHER		3

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL124Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-2-29	AN960C10		. . . TERM. BLK WASHER		2
5-2-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1092-22		. . NAME PLATE		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	AN500A8-8		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064		. ARMATURE ASSY		1
5-1-45	200SGL1107		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-54	150SG1014		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL124Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10XL2		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL129Q
200SGL129Q(1)
200SGL129Q(3)
200SGL129Q(4)
200SGL129Q(5)
200SGL129Q(8)

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL129Q	Shunt	200SGL1083-2	AS972	31-Oct-94

Installations

Type Certificates

Valmont
Agusta

L-90-TP Rdigo
Koala

Supplemental Type Certificates

None

Disassembly and Assembly

Speed Sensor hookup:

Uses magnetic pickup and spur gear
Black lead = A White lead = B

Testing

Shunt field terminal A - D resistance tolerance:

1.80 - 2.06 Ohms

PC Board Capacitance tolerance:

.33 MFD +/- 10%

Terminal block Capacitance tolerance:

B-E
E-Grnd
B-Grnd.

Hi-Pot Test (Stator)

250 VAC for 1 second
Max leakage current allowable: 2.5 milliamps
A-Grnd., A-B, A-E
B-Grnd., B-E
E-Grnd.

Equalizing Voltage Chart

Chart 2

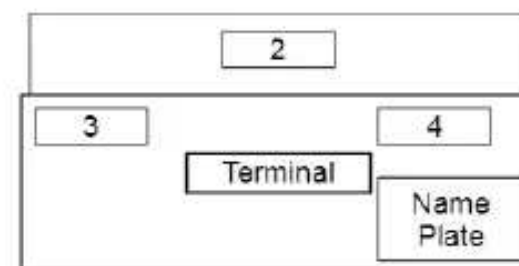
Modified Status

Applicable Decals

Decals 3

Fan End (Print Reversed)

- | | |
|---|----------------|
| 2 | Hi-Pot Warning |
| 3 | APC XL2 Label |
| 4 | QAD Warning |



Other Applicable Technical Data:

1,000 Hr TBO with 500 Hr Mid-life brush change

Part Differences by Model

Model	Data Plate	QAD Kit	Mounting Flange	Stator Assy.
200SGL129Q-(1)	200SGL1092-24	200SGL1083-2	200SGL1040-2	200SGL1156-6
200SGL129Q-(3)	200SGL1092-25	200SGL1083-3	200SGL1040-3	200SGL1156-6
200SGL129Q-(4)	200SGL1092-26	200SGL1083-3	200SGL1040-3	200SGL1156-6
200SGL129Q-(5)	200SGL1092-27	200SGL1083-3	200SGL1040-3	200SGL1156-6
200SGL129Q-(8)	200SGL1092-39	200SGL1083-3	200SGL1040-3	200SGL1156-4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL129Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL129Q		STARTER GENERATOR		RF
5-1-	200SGL1083-2		. QAD KIT [200SGL129Q(1)]		1
5-1-	200SGL1083-3		. QAD KIT [200SGL129Q(3), (4), (5), and (8)]		1
5-1-1	200SGL1040-2		. . MOUNTING FLANGE [200SGL129Q(1)]		1
5-1-1	200SGL1040-2		. . MOUNTING FLANGE [200SGL129Q(3), (4), (5), and (8)]		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-B		. BALL BEARING, DRIVE END		1
5-1-6A	150SG1052-1-B		. BALL BEARING, A DRIVE END		1
5-1-7	200SGL1103-9		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-2		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-2		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		6
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL129Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-18	200SGL1219-1		. SPUR GEAR		1
5-2-	200SGL1126-5		. STATOR & TERM. BLOCK ASSY		1
5-2-19C	200SGL1156-4		. . STATOR ASSY [200SGL129Q(8)		1
5-2-19D	200SGL1156-6		. . STATOR ASSY [200SGL129Q(1), (3), (4), and (5)		1
5-2-22	200SGL1061-3		. . AXIAL TERMINAL BLOCK		1
5-2-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		2
5-2-27	AN960C616		. . . TERM. BLK WASHER		2
5-2-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-2-29	AN960C10		. . . TERM. BLK WASHER		2
5-2-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-2-33	MS3112E8-3P		. . CONNECTOR		1
5-2-34	200SG1221		. . CONNECTOR SPACER		1
5-2-35	300SG1222-2		. . MAGNETIC PICK-UP		1
5-1-36	MS35265-19		. . CONNECTOR SCREW		4
5-1-37	200SGL1092-24		. . NAME PLATE [200SGL129Q(1)]		1
5-1-37	200SGL1092-25		. . NAME PLATE [200SGL129Q(3)]		1
5-1-37	200SGL1092-26		. . NAME PLATE [200SGL129Q(4)]		1
5-1-37	200SGL1092-27		. . NAME PLATE [200SGL129Q(5)]		1
5-1-37	200SGL1092-39		. . NAME PLATE [200SGL129Q(8)]		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL129Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006-1		. BEARING RETAINER		1
5-1-43	MS24694-S5		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064-5-XL		. ARMATURE ASSY, [200SGL129Q(1) and (5) Only]		1
5-1-45	200SGL1107		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-54	200SGL1011		. FAN COVER ASSY [200SGL129Q(1) and (5) Only]		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL129Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-3		. . "XL" BRUSH [200SGL129Q(1) and (5) Only]		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL130Q
200SGL130Q-2
200SGL130Q-3

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL130Q	Series	200SGL1083	Wet	26-Feb-91

Installations

Type Certificates

-2	Eurocopter	AS 350	Approved 11 Dec., 2003
-3	Eurocopter	EC 135	
	Eurocopter	EC 145	

Supplemental Type Certificates

None

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance: 1.80 - 2.06 Ohms

PC Board Capacitance tolerance: .33 MFD +/- 10%

Terminal block Capacitance tolerance:
B-E
E-Grnd
B-Grnd.

Hi-Pot Test (Stator)

250 VAC for 1 second
Max leakage current allowable: 2.5 milliamps
A-Grnd., A-B, A-E
B-Grnd., B-E
E-Grnd.

Equalizing Voltage Chart

Chart 1

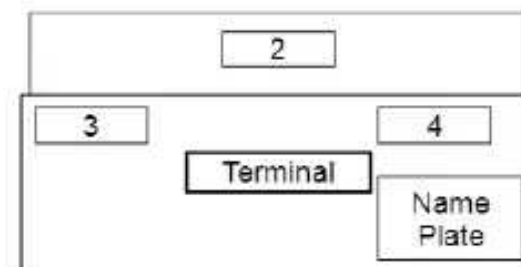
Modified Status

Applicable Decals

Decals 3

Fan End (Print Reversed)

- 2 Hi-Pot Warning
- 3 APC XL2 Label
- 4 QAD Warning



Other Applicable Technical Data:

1,000 Hr TBO with 500 Hr Mid-life brush change

Part Differences by Model

<u>Model</u>	<u>Data Plate</u>	<u>QAD Kit</u>	<u>Air Inlet Cover</u>
200SGL130Q-2	200SGL1092-54	200SGL1083-1	200SGL1014-15
200SGL130Q-3	200SGL1092-55	200SGL1083	200SGL1011

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL130Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL130Q		STARTER GENERATOR		RF
5-1-	200SGL1083		. QAD KIT [200SGL130Q and Q-3]		1
5-1-	200SGL1083-1		. QAD KIT [200SGL130Q-2]		1
5-1-1	200SGL1040-5		. . MOUNTING FLANGE		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-B		. BALL BEARINGS		2
5-1-7	200SGL1103-10		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-3		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-3		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		6
5-1-15	150SG1106-1		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	250SG1113		. FLAT WASHER		8
5-2-	200SGL1126		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156		. . STATOR ASSY		1
5-2-22	200SGL1061		. . AXIAL TERMINAL BLOCK		1
5-2-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		3
5-2-27	AN960C616		. . . TERM. BLK WASHER		3

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL130Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-2-29	AN960C10		. . . TERM. BLK WASHER		2
5-2-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1092-17		. . NAME PLATE [200SGL130Q]		1
5-1-37	200SGL1092-54		. . NAME PLATE [200SGL130Q-2]		1
5-1-37	200SGL1092-55		. . NAME PLATE [200SGL130Q-3]		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	AN500A8-8		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064-XL		. ARMATURE ASSY		1
5-1-45	200SGL1107		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-54	200SGL1014		. FAN COVER ASSY [200SGL130Q]		1

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL130Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1-54	200SGL1014-15		. FAN COVER ASSY [200SGL130Q-2]		1
5-1-54	200SGL1011		. FAN COVER ASSY [200SGL130Q-3]		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10XL2		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL132Q

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL132Q	Series	200SGL1083	Wet	24-Mar-93

Installations

Type Certificates

Bell Hellicopter- 206L-1, 206L-3

Supplemental Type Certificates

SH123NE Bell- 206L-1 10 February, 1983
(Amended 3 May, 1993)
SH401NE Bell- Textron- 206L-3 19 November, 1985
(Amended 24 March, 1993)

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance:

1.80 - 2.06 Ohms

PC Board Capacitance tolerance:

.33 MFD +/- 10%

Terminal block Capacitance tolerance:

B - E
B - C
E - Grnd.
B-Grnd.
C-Grnd.
C - E

Hi-Pot Test (Stator)

250 VAC for 1 second
Max leakage current allowable: 2.5 milliamps
A-Grnd., A-B, A-E, A-C
B-Grnd., B-E
C-Grnd.
E-Grnd.

Equalizing Voltage Chart

Chart 1

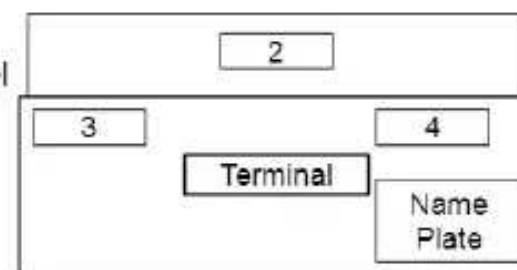
Modified Status

Applicable Decals

Decals 3

Fan End (Print Reversed)

- 2 Hi-Pot Warning
- 3 APC XL2 or XL Label
- 4 QAD Warning



Other Applicable Technical Data:

SB150SG128-1

XL2 Brushes

Shorter than 200SGL111Q and with beveled fan cover. Better fit choice for Bell installations

Bell name plated units do not use APC XL Label

Bell units not stamped XL2 behind S/N

Bell name plate 200SGL1092-59 for 200SGL132Q-3

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL132Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL132Q		STARTER GENERATOR		RF
5-1-	200SGL1083		. QAD KIT		1
5-1-1	200SGL1040		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-1		. BALL BEARINGS		2
5-1-7	200SGL1103-11		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-1		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-1		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		4
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-	200SGL1126-6		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156		. . STATOR ASSY		1
5-2-22	200SGL1061-4		. . AXIAL TERMINAL BLOCK		1
5-2-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		3
5-2-27	AN960C616		. . . TERM. BLK WASHER		3

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL132Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-2-29	AN960C10		. . . TERM. BLK WASHER		2
5-2-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1092-29		. . NAME PLATE		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	MS35265-45		. BEARING RETAINER SCREW		4
4-1-44	200SGL1064		. ARMATURE ASSY		1
5-1-45	200SGL1107-2		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-54	200SGL1014-5		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL132Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10XL2		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL132Q-4

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL132Q-4	series	200SGL1083	Wet	12-May-09

Installations

Type Certificates

Supplemental Type Certificates

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance:

1.80 to 2.06 Ohms

PC Board Capacitance tolerance:

.33 MFD +/- 10%

Terminal block Capacitance tolerance:

B-E -1.8 to -2.2 UFD

B-C -.015 to -.0238 UFD

E-Grnd. -.015 to -.0235 UFD

B-Grnd. -.016 to -.0240 UFD

C-Grnd. -1.8 to -2.2 UFD

C-E -.015 to -.236 UFD

Hi-Pot Test (Stator)

250 VAC for 1 second

Max leakage current allowable: 2.5 milliamps

A-Grnd., A-B, A-E, A-C

B-Grnd., B-E

C-Grnd.

E-Grnd.

Equalizing Voltage Chart

Chart 2

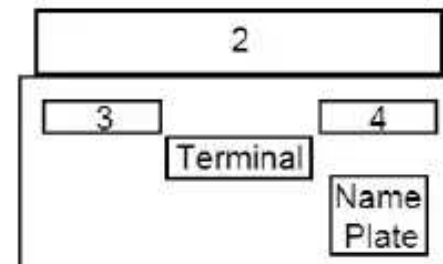
Modified Status

Applicable Decals

Decals 3

Fan End (Print Reversed)

- 2 Hi-Pot Warning
- 3 APC XL2 Label
- 4 QAD Warning



Other Applicable Technical Data:

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL132Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL111Q		STARTER GENERATOR		RF
5-1-	200SGL1083		. QAD KIT		1
5-1-1	200SGL1040		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-1		. BALL BEARINGS		2
5-1-7	200SGL1103-11		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-1		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-1		. . DRIVE END BELL		1
5-1-11	150SG1005-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		4
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-	200SGL1126		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156		. . STATOR ASSY		1
5-1-20	200SGL1061-4		. . AXIAL TERMINAL BLOCK		1
5-1-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		3
5-1-27	AN960C616		. . . TERM. BLK WASHER		3

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL132Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-1-29	AN960C10		. . . TERM. BLK WASHER		2
5-1-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-1-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-1-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1094-64		. . NAME PLATE		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	MS35265-45		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064		. ARMATURE ASSY		1
5-1-45	200SGL1107-5		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153-2		. COMM HEAD ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-52	200SGL1011-17		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL132Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-3-1	200SGL1154	.	. . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10		. . BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL144Q

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL144Q	series	200SGL1083	Wet	14-Jul-98

Installations

Type Certificates Sikorsky S76 C+

Supplemental Type Certificates None

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance: 1.8 to 2.06 Ohms
PC Board Capacitance tolerance: .33 MFD +/- 10%
Terminal block Capacitance tolerance: B-E -1.8 to -2.2 UFD
B-C -.015 to -.0238 UFD
E-Grnd. -.015 to -.0235 UFD
B-Grnd. -.016 to -.0240 UFD
C-Grnd. -1.8 to -2.2 UFD
C-E -.015 to -.236 UFD

Hi-Pot Test (Stator) 250 VAC for 1 second
Max leakage current allowable: 2.5 milliamps
A-Grnd., A-B, A-E, A-C
B-Grnd., B-E
C-Grnd.
E-Grnd.

Equalizing Voltage Chart Chart 1

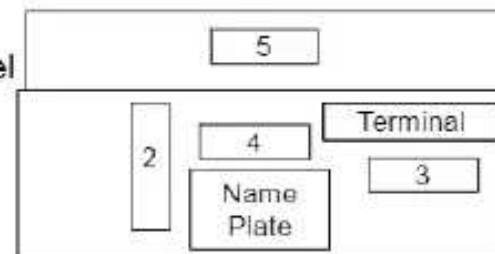
Modified Status

Applicable Decals

Decals 3

Fan End (Print Reversed)

- 2 Hi-Pot Warning
- 3 APC XL OR XL2 Label
- 4 QAD Warning
- 5 Fan Cover

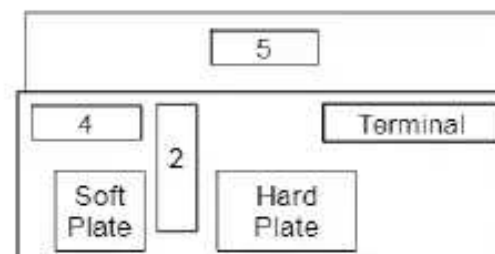


Other Applicable Technical Data:

SB150SG128-1 XL2 Brushes

Sikorsky does not use XL2 brushes **Sikorsky Decal Placement>>**

Other customers can use XL2 brushes, however:
Sikorsky P/N & Name must be removed as well as
soft nameplate will not be used.



AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL144Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL144Q		STARTER GENERATOR		RF
5-1-	200SGL1083		. QAD KIT		1
5-1-1	200SGL1040		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-10		. BALL BEARINGS		2
5-1-7	200SGL1103-13		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-1		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-1		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		4
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-	200SGL1126		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156		. . STATOR ASSY		1
5-2-22	200SGL1061		. . AXIAL TERMINAL BLOCK		1
5-2-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		3
5-2-27	AN960C616		. . . TERM. BLK WASHER		3

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL144Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-2-29	AN960C10		. . . TERM. BLK WASHER		2
5-2-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1092-43		. . NAME PLATE		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-2-40B	200SGL1381		. . FAN COVER DECAL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	MS35265-45		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064		. ARMATURE ASSY		1
5-1-45	200SGL1107-2		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-2-52A 5-2-57 5-2-58	200SGL1014-8		. AIR INLET KIT ASSY		1

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL144Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1-54	200SGL1014-6		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-2-58	200SGL1014-7		. RADIAL AIR INLET		1
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10XL2		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL145Q

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL145Q	Series	200SGL1083	Wet	14-Jul-98

Installations

Type Certificates Piper- Meridian

Supplemental Type Certificates None

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance: 1.80 - 2.06 Ohms
PC Board Capacitance tolerance: .33 MFD +/- 10%
Terminal block Capacitance tolerance:
B - E
B - C
E - Grnd
B-Grnd.
C-Grnd.
C - E
Hi-Pot Test (Stator) 250 VAC for 1 second
Max leakage current allowable: 2.5 milliamps
A-Grnd., A-B, A-E, A-C
B-Grnd., B-E
C-Grnd.
E-Grnd.

Equalizing Voltage Chart Chart 1

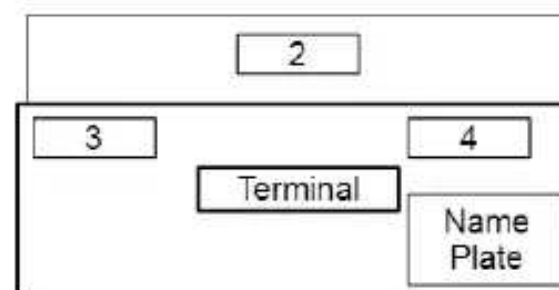
Modified Status

Applicable Decals

Decals 3

Fan End (Print Reversed)

- 2 Hi-Pot Warning
- 3 APC XL2 Label
- 4 QAD Warning



Other Applicable Technical Data:

SB150SG128-1 XL2 Brushes

Check for non-XL series

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL145Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL145Q		STARTER GENERATOR		RF
5-1-	200SGL1083		. QAD KIT		1
5-1-1	200SGL1040		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-B		. BALL BEARINGS		2
5-1-7	200SGL1103		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-1		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-1		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		4
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-	200SGL1126		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156		. . STATOR ASSY		1
5-2-22	200SGL1061		. . AXIAL TERMINAL BLOCK		1
5-2-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		3
5-2-27	AN960C616		. . . TERM. BLK WASHER		3

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL145Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-28	SPS27FT-1032		 TERM. BLK NUT (V56878)		2
5-2-29	AN960C10		 TERM. BLK WASHER		2
5-2-30	MS35265-65		. . . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1092-47		. . . NAME PLATE		1
5-1-38	MS21318-13		. . . NAME PLATE SCREW		4
5-1-39	150SG1139		. . . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	MS35265-45		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064		. ARMATURE ASSY		1
5-1-45	200SGL1107		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-55	250SG1116		. FAN COVER SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL145Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-58	200SGL1011-8		. RADIAL AIR INLET		1
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009		. . STANDARD BRUSH		4
5-3-11A	200SGL1009-10XL2		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL147Q-2
200SGL147Q-3

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL147Q-2	Shunt	200SGL1083-2	Wet	17-Aug-04
200SGL147Q-3	Shunt	200SGL1083-2	Wet	15-Oct-07

Installations

Type Certificates

Agusta A109 Power, A109 LUH

Supplemental Type Certificates

None

Disassembly and Assembly

Speed Sensor hookup:

Uses magnetic pickup and spur gear
Black lead = A White lead = B

Testing

Shunt field terminal A - D resistance tolerance:

1.80 - 2.06 Ohms

PC Board Capacitance tolerance:

.33 MFD +/- 10%

Terminal block Capacitance tolerance:

B - E

E - Grnd

B-Grnd.

Hi-Pot Test (Stator)

250 VAC for 1 second

Max leakage current allowable: 2.5 milliamps

A-Grnd., A-B, A-E

B-Grnd., B-E

E-Grnd.

Equalizing Voltage Chart

Chart 2

Modified Status

Applicable Decals

Decals 3

Fan End (Print Reversed)

2

Hi-Pot Warning

3

APC XL2 Label

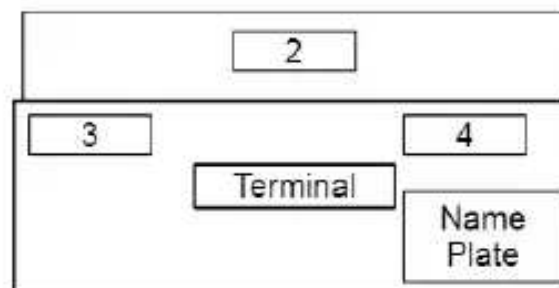
4

QAD Warning

Other Applicable Technical Data:

P/N status: -3

Description	Date	Change	Stamp
TSO	17-Aug-04	Fan Cover	-2
SB200SG140	5-Dec-07	Bearings	-3



AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL147Q-2, -3

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL147Q-2, -3		STARTER GENERATOR		RF
5-1-	200SGL1083-2		. QAD KIT		1
5-1-1	200SGL1040-2		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-3	M83248-1-113		. O-RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. GREASE SLINGER		2
5-1-6	200SGL1052-1-B		. BALL BEARINGS, DRIVE END [200SGL147Q-2]		1
5-1-6	200SGL1052-5		. BALL BEARING, DRIVE END [200SGL147Q-3]		1
5-1-6A	150SG1052-B		. BALL BEARINGS, A DRIVE END [200SGL147Q-2]		1
5-1-6A	150SG1052-1		BALL BEARING, A DRIVE END [200SGL147Q-3]		1
5-1-7	200SGL1103-18		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-2		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-2		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		6
5-2-14	MS35489-1		. . GROMMET		1
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

4-2. Detailed Parts List - 200SGL147Q-2, -3 (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1-17	150SG1113		. FLAT WASHER		8
5-2-18	200SGL1219-1		. SPUR GEAR		1
5-2-19	200SGL1126-5		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156-6		. . STATOR ASSY		1
5-2-22	200SGL1061-3		. . AXIAL TERMINAL BLOCK		1
5-2-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		2
5-2-27	AN960C616		. . . TERM. BLK WASHER		2
5-2-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2
5-2-29	AN960C10		. . . TERM. BLK WASHER		2
5-2-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-2-33	MS112E8-3P		. . CONNECTOR		1
5-2-34	200SG1221		. . SPACER CONNECTOR		1
5-2-35	300SG1222-2		. . MAGNETIC PICK-UP		1
5-2-36	MS35265-19		. . SCREW CONNECTOR		4
5-1-37	200SGL1092-57		. . NAME PLATE		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006-1		. BEARING RETAINER		1
5-1-43	MS24694-S5		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064-5-XL		. ARMATURE ASSY		1

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL147Q-2, -3 (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1-45	200SGL1107		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-54	200SGL1011-15		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-3-	200SGL1138		. . HOLDER ASSY., BRUSH & SPRING		4
5-3-1	200SGL1154		. . . BRUSH HOLDER		4
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10XL2		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Starter Generator
200SGL150Q
200SGL150Q-1
200SGL150Q-2

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

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AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL150Q	Series	200SGL1083-11	Wet	28-Mar-06

Installations

Type Certificates Bell- 417

Supplemental Type Certificates None

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance: 1.80 T 2.06 ohms

PC Board Capacitance tolerance: .33 MFD +/- 10%

Terminal block Capacitance tolerance:
B to E
E to Grnd.
B to Grnd.

Hi-Pot Test (Stator) 250 VAC for 1 second
Max leakage current allowable: 2.5 milliamps
A-Grnd., A-B, A-E
B-Grnd., B-E
E-Grnd.

Equalizing Voltage Chart Chart 2

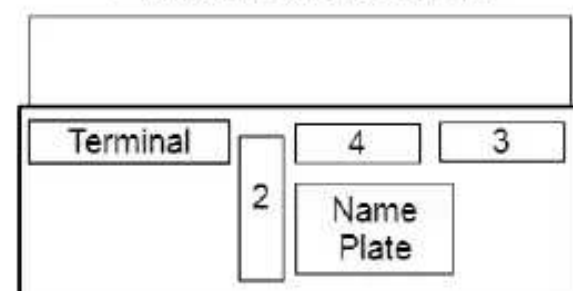
Modified Status

Applicable Decals

Decals 3

- 2 Hi-Pot Warning
- 3 APC XL2 Label
- 4 QAD Warning

Fan End (Print Reversed)



Other Applicable Technical Data:

SB150SG128-1XL2 Brushes

Description	Date	Change	Stamp
Drawing	28-Mar-06	Air Inlet	-2
Drawing	28-Mar-06	Inlet & QAD	-1
TSO	28-Mar-06	No inlet-QAD	Basic

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
200SGL SERIES STARTER GENERATOR

5-2. Detailed Parts List - 200SGL150Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL150Q		STARTER GENERATOR		RF
5-1-	200SGL1083-11		. QAD KIT		1
5-1-1	200SGL1040-11		. . MOUNTING FLANGE		1
5-1-2	150SG1108		. . CLAMP RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-1		. BALL BEARINGS		2
5-1-7	200SGL1103-11		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-3		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-3		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		4
5-1-13	MS21318-14		. . DRIVE SCREW		4
5-1-15	150SG1106		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-	200SGL1126-4		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156		. . STATOR ASSY		1
5-2-22	200SGL1061-4		. . AXIAL TERMINAL BLOCK		1
5-2-26	SPS27FT-624		. . TERM. BLK. NUT (V56878)		3
5-2-27	AN960C616		. . . TERM. BLK WASHER		3
5-2-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2

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5-2. Detailed Parts List - 200SGL150Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-29	AN960C10		 TERM. BLK WASHER		2
5-2-30	MS35265-65		. . . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1092-62		. . . NAME PLATE		1
5-1-38	MS21318-13		. . . NAME PLATE SCREW		4
5-1-39	150SG1139		. . . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	MS35265-45		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064-XL		. ARMATURE ASSY		1
5-1-45	200SGL1107-2		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153-2		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-54	200SGL1014-16		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-3-	200SGL1138-1		. . . HOLDER ASSY., BRUSH & SPRING		4
5-3-1	200SGL1154-3		. . . BRUSH HOLDER		4

AIRCRAFT PARTS CORP.
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5-2. Detailed Parts List - 200SGL150Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-3-2	200SGL1005		... SPRING		8
5-3-3	200SGL1057		... P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		... COMM HEAD		1
5-3-5	150SG1112-1		... BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		... PLAIN WASHER		16
5-3-7	150SG1045		... TUBE INSULATOR		8
5-3-8	150SG1046		... BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		... INSULATOR WASHER		4
5-3-10	SPS27FT-1032		... SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10XL2		... "XL2" BRUSH		4
5-3-12	200SG1020		... BRUSH MTG SCREW		4

AIRCRAFT PARTS CORP.
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Starter Generator
200SGL151Q

Overhaul Data Sheet
Detailed Parts List

AIRCRAFT PARTS CORP.
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AIRCRAFT PARTS CORP.
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Overhaul Data Sheet

Model	Starting Type	QAD Kit	Drive Spline	Approval Status
200SGL151Q	Shunt	200SGL1083-9	Dry	28-Mar-06

Installations

Type Certificates Bell- 417/ARH

Supplemental Type Certificates None

Disassembly and Assembly

Testing

Shunt field terminal A - D resistance tolerance: 1.80 T 2.06 ohms

PC Board Capacitance tolerance: .33 MFD +/- 10%

Terminal block Capacitance tolerance:
B to E
E to Grnd.
B to Grnd.

Hi-Pot Test (Stator) 250 VAC for 1 second
Max leakage current allowable: 2.5 milliamps
A-Grnd., A-B, A-E
B-Grnd., B-E
E-Grnd.

Equalizing Voltage Chart Chart 2

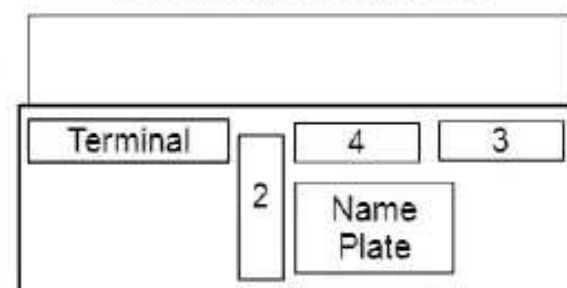
Modified Status

Applicable Decals

Decals 3

2 Hi-Pot Warning
3 APC XL2 Label
4 QAD Warning

Fan End (Print Reversed)



Other Applicable Technical Data:

SB150SG128-1XL2 Brushes

AIRCRAFT PARTS CORP.
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5-2. Detailed Parts List - 200SGL151Q

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-1	200SGL151Q		STARTER GENERATOR		RF
5-1-	200SGL1083-11		. QAD KIT		1
5-1-1	200SGL1040-10		. . MOUNTING FLANGE		1
5-1-2	200SGL1108-1		. . CLAMP RING		1
5-1-4	MS16624-4066		. RETAINING RING		1
5-1-5	150SG1017		. BAFFLE DISC		2
5-1-6	150SG1052-1		. BALL BEARINGS		2
5-1-7	200SGL1103-21		. DRIVE SHAFT ASSY		1
5-1-8	150SG1070		. FRICTION RING		1
5-1-9	150SG1073		. DAMPENER PLATE		1
5-1-	200SGL1301-3		. DE HOUSING & SCREEN ASSY.		1
5-1-10	200SGL1028-3		. . DRIVE END BELL		1
5-1-11	150SG1004-1		. . SCREEN		1
5-1-12	250SG1302		. . SCREEN WASHER		6
5-1-13	MS21318-14		. . DRIVE SCREW		4
5-1-15	150SG1106-1		. MTG HEAD SCREW		8
5-1-16	150SG1379-8		. LOCK WASHER		8
5-1-17	150SG1113		. FLAT WASHER		8
5-2-	200SGL1126-3		. STATOR & TERM. BLOCK ASSY		1
5-2-19D	200SGL1156-1		. . STATOR ASSY		1
5-2-22	200SGL1061-3		. . AXIAL TERMINAL BLOCK		1
5-2-26	SPS27FT-624		. . . TERM. BLK. NUT (V56878)		3
5-2-27	AN960C616		. . . TERM. BLK WASHER		3
5-2-28	SPS27FT-1032		. . . TERM. BLK NUT (V56878)		2

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
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5-2. Detailed Parts List - 200SGL151Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-2-29	AN960C10		. . . TERM. BLK WASHER		2
5-2-30	MS35265-65		. . TERM. BLK. MTG. SCREW		2
5-2-31	MS35338-43		. . TERM. BLK. LOCK WASHER		2
5-2-32	AN960-10L		. . TERM. BLK. FLAT WASHER		2
5-1-37	200SGL1092-63		. . NAME PLATE		1
5-1-38	MS21318-13		. . NAME PLATE SCREW		4
5-1-39	150SG1139		. . HI-POT WARNING DECAL		1
5-1-40	150SG1168		. . QAD WARNING DECAL		1
5-1-40A	200SGL1360-1		. . APC XL2 LABEL		1
5-1-41	200SGL1006		. BEARING RETAINER		1
5-1-42	150SG1379-8		. BEARING RETAINER LOCK WASHERS		4
5-1-43	MS35265-45		. BEARING RETAINER SCREW		4
5-1-44	200SGL1064-XL		. ARMATURE ASSY		1
5-1-45	200SGL1107-2		. FAN ASSY		1
5-1-46	200SGL1093		. BELLEVILLE SPRINGS		2
5-1-47	150SG1071		. SELF LOCKING NUT		1
5-1-48	200SGL1153-2		. COMM END BELL ASSY		1
5-1-49	150SG1105		. COMM. HEAD MTG SCREW		8
5-1-50	150SG1379-8		. COMM. HEAD MTG LOCK WASHER		8
5-1-51	250SG1113		. COMM. HEAD MTG. FLAT WASHER		8
5-1-54	200SGL1014-18		. FAN COVER ASSY		1
5-1-55	250SG1116		. FAN COVER SCREW		4
5-3-	200SGL1138-1		. . HOLDER ASSY., BRUSH & SPRING		4
5-3-1	200SGL1154-3		. . . BRUSH HOLDER		4

AIRCRAFT PARTS CORP.
OVERHAUL MANUAL WITH IPL
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5-2. Detailed Parts List - 200SGL151Q (cont.)

FIG & ITEM	PART NUMBER	AIRLINE PART NO.	1 2 3 4 5 6 7 NOMENCLATURE	EFF CODE	UNITS PER ASSY
5-3-2	200SGL1005		. . . SPRING		8
5-3-3	200SGL1057		. . P.C. BOARD ASSY.		2
5-3-4	200SGL1008-1		. . COMM HEAD		1
5-3-5	150SG1112-1		. . BRUSH HOLDER SCREW		8
5-3-6	AN960-10L		. . PLAIN WASHER		16
5-3-7	150SG1045		. . TUBE INSULATOR		8
5-3-8	150SG1046		. . BRUSH BOX INSULATOR WASHER		8
5-3-9	150SG1114		. . INSULATOR WASHER		4
5-3-10	SPS27FT-1032		. . SELF LOCKING NUT (V56878)		8
5-3-11	200SGL1009-10XL2		. . "XL2" BRUSH		4
5-3-12	200SG1020		. . BRUSH MTG SCREW		4

